Return Concerned Adaptive Control of Product Offerings

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Abstract. Problem definition: Product returns are a major operational challenge in online retail, where lenient return policies and post-purchase uncertainty generate both substantial costs and delayed information. We study a dynamic assortment optimization problem in which a retailer must learn customer purchase and return behavior over time while return outcomes are observed with delay and only within a finite return window. The problem is motivated by a collaboration with a major European fast-fashion retailer.

Methodology/results: We develop a unified two-stage purchase–return model in which the purchase decision and the subsequent keep/return decision are generated from shared latent primitives. To solve the resulting learning problem, we propose COBRA (*Confidence Bounds for Return-Aware Assortments*), an epoch-based algorithm that combines product-level likelihood estimation, learning of the in-window return-delay profile, and optimistic assortment selection under joint uncertainty. We establish a regret bound that makes explicit the effects of delayed and window-censored returns, and we complement it with a lower bound showing that partial return observability fundamentally affects the learning problem. Numerical experiments on both synthetic instances and real retail data show that incorporating return information can substantially improve profit performance relative to return-blind approaches.

Managerial implications: Our results show that returns should be treated not only as a cost, but also as a delayed information source for merchandising decisions. In the tested settings, return-aware learning improves assortment quality and often yields lower realized return volumes as a secondary operational benefit. The analysis also highlights a managerial tradeoff in return-window design: expanding the window makes more return information observable, but it also increases the temporal difficulty of learning from recent purchases.

Key words: Dynamic assortment optimization, Product returns, Delayed feedback, Online learning

1. Introduction

In recent years, the retail industry has witnessed a profound transformation in consumer behavior, notably marked by a significant uptick in product returns. This shift has been catalyzed by several factors, chief among them being the proliferation of online channels and the widespread adoption of lenient return policies by retailers. While online shopping has revolutionized the way people shop by offering convenience and comfort, it also comes with a drawback: returns Chevalier (2023). The dangerous news for retailers is that around 20% to 30% of products ordered online are returned, as compared to only 9% in brick-and-mortar stores Dopson (2023).

The surge in returned goods poses substantial challenges for retailers both economically and environmentally. To contextualize this trend, consider the staggering figures from the 2023 NRF report on the consumer returns in the retail Industry NRF (2023): American consumers returned a whopping $743 billion worth of products, representing a remarkable 68% increase from just three years prior in 2020 NRF (2020). This surge not only impacts retailers' bottom lines but also raises significant environmental concerns, with an estimated 10% of returns ultimately ending up in landfills.

Despite the escalating volume of returns, the retail industry has largely overlooked the management of this phenomenon. Many retailers perceive high return rates as an inevitable byproduct of their operations, viewing generous return policies as indispensable for expanding their market share and fostering consumer loyalty. Moreover, the advent of online channels and the relaxation of return restrictions, particularly amid the COVID-19 pandemic, have further entrenched consumers' perception of purchases as virtually risk-free product experimentation rather than as conclusive shopping journeys.

This leniency in return practices is not confined to the United States alone but extends globally, with over 50% of retailers in the European Union also embracing free return shipping policies Gowans (2022). While some retailers have experimented with strategies to mitigate returns, such as leveraging consumer reviews and high-resolution images to aid in product selection, the broader integration of returns data into the retail ecosystem remains largely unexplored. Only a fraction of retailers actively incorporate returns data into their merchandising and product development processes, thereby missing out on valuable opportunities to optimize their assortment decisions and enhance overall operational efficiency.

In collaboration with one of Europe's largest online fast-fashion retailers, we study how returns data can be incorporated into dynamic assortment decisions. Our industrial partner launches a large number of new products each season–for example, roughly 30,000 items in a typical Winter/Spring season–many of which have little or no prior sales history. In this setting, the retailer must learn purchase and return behavior jointly while updating assortments within season. Even with consumer reviews and high-resolution images, customers retain uncertainty about product fit and post-purchase satisfaction. Combined with the retailer's 30-day return window, this generates delayed return feedback and makes timely learning particularly challenging.

Recognizing the pivotal role of returns data in shaping product assortment decisions, this paper seeks to address the following research question: in light of delayed returns data, *how can retailers best utilize this information to enhance their product assortment decisions and minimize their return costs?* By addressing this question, we aim to provide insights and recommendations that enable retailers to capitalize on returns data, particularly the constraints it imposes on retailers' ability to react and adapt their assortment strategies in a timely manner.

Main Contributions: Our main results and contributions can be summarized as follows.

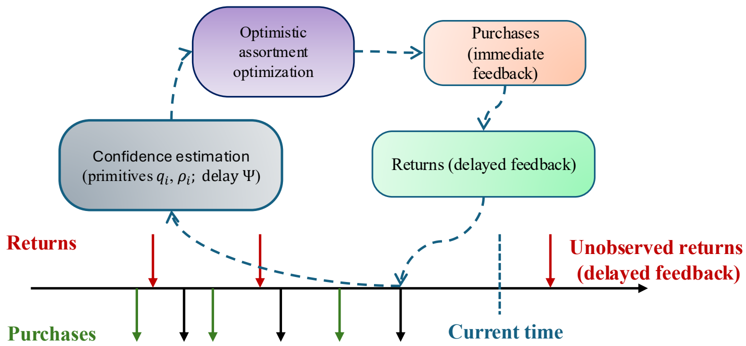

Figure 1 **Overview of the COBRA algorithm. In each epoch the algorithm constructs confidence sets for the latent primitives governing purchase and return behavior, selects an optimistic assortment under these confidence sets, and then collects purchase and delayed return observations until a no-purchase event triggers the next update. Returns that occur beyond the return window remain unobserved.**

1. **A dynamic assortment learning model with delayed and censored returns.** We formulate a dynamic assortment optimization problem in which purchase and return behavior are jointly governed by latent utility primitives. In our framework, purchase decisions follow a multinomial logit model, while post-purchase return decisions arise from a stochastic comparison between realized product satisfaction and return hassle. Returns are observed only with delay and only within a finite return window, creating both delayed and censored feedback. This structure captures a key operational feature of modern online retail settings where return information arrives gradually and may remain partially unobserved.
2. **The COBRA algorithm for return-aware assortment learning.** To address the resulting exploration–exploitation tradeoff, we develop an epoch-based learning algorithm called **COBRA** (*Confidence Bounds for Return-Aware Assortments*). The algorithm maintains confidence regions for the latent primitives governing purchase and return behavior together with estimates of the return-delay profile. In each epoch, COBRA selects an optimistic assortment that maximizes the retailer's profit objective under these confidence sets, and then collects purchase and return observations until a no-purchase event triggers the next update. Figure 1 provides a schematic overview of the learning-and-optimization cycle implemented by the algorithm.
3. **Performance guarantees.** We establish a regret bound for **COBRA** that makes explicit the effects of delayed and window-censored returns. In particular, the bound identifies both the statistical difficulty created by partial in-window return observability and the censoring effect created by unresolved recent purchases. We also derive a lower bound showing that the leading $\sqrt{NT}$ dependence is unavoidable and that the return window affects the fundamental difficulty of learning through the fraction of returns that are observable within the window.

4. **Operational insights from numerical experiments.** Finally, we evaluate the proposed approach through numerical experiments using both synthetic data and a real dataset obtained from our industrial partner. The experiments illustrate the value of incorporating delayed return information into assortment learning and quantify how delayed feedback and the return window affect learning performance and show the value of incorporating return information into dynamic assortment decisions.

Organization: The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model formulation. Section 4 develops the COBRA algorithm and its regret guarantee. Section 5 presents an efficient solution method for the optimistic assortment subproblem. Section 6 provides the regret lower bound. Section 7 reports the numerical experiments and managerial insights. Section 8 concludes.

2. Literature Review

Our paper falls within the general theme of assortment optimization. To the best of our knowledge, ours is the first paper to study dynamic assortment optimization with jointly learned purchase and return behavior under delayed and window-censored return feedback. There are three closely related streams of literature: dynamic assortment learning, operations models with product returns, and online learning with delayed feedback.

Assortment optimization. Assortment optimization studies how a firm should choose the set of products offered to customers under a model of choice behavior. While a variety of choice models have been used in this literature, including nested-logit(see e.g., Gallego and Topaloglu (2014), Li et al. (2015)), Markov-chain (see e.g., Blanchet et al. (2016)), and nonparametric models(see e.g., Farias et al. (2013)), the multinomial logit (MNL) model remains one of the most tractable and widely used frameworks (Luce 1960, Plackett 1975). In the unconstrained case, Talluri and Van Ryzin (2004) showed that revenue-ordered assortments are optimal, while constrained settings are substantially more complex (Désir et al. 2014). See Kök et al. (2015) for a broader overview.

In many practical situations, the parameters of the choice model, such as customer preferences or product attractiveness, are unknown and need to be learned from data. Until today, different learning approaches have been proposed in the literature. Caro and Gallien (2007) modeled fast fashion assortments as a multi-armed bandit model with the assumption that product demands are independent. In terms of the MNL choice model, Rusmevichientong et al. (2010) and Sauré and Zeevi (2013) utilized an explore-then-exploit approach for learning the attraction parameters of the MNL model. Agrawal et al. (2017) employed Thompson sampling, while Agrawal et al. (2019) applied an upper confidence bound approach. Chen et al. (2021) developed a tri-section-based policy for the potential function derived from an unconstrained assortment problem, while Chen et al. (2023) introduced an adversarial elimination method for the assortment problem when a portion of customers are outliers.

Our paper contributes to this literature by introducing delayed and window-censored product returns into dynamic assortment learning, and by modeling purchase and return behavior through shared latent primitives rather than as unrelated reduced-form objects.

Product returns. The literature on product returns encompasses a range of studies addressing the impact of returns on assortment optimization, inventory management, and pricing strategies.

A significant portion of the literature focused on mitigating the impact of returns on inventory decisions. Early works focused on deterministic models, such as those by Schrady (1967) and Fleischmann et al. (1997), while stochastic models, exemplified by Heyman (1977) and Fleischmann et al. (2002), introduced uncertainty considerations. Simplifying assumptions regarding returns are common in this literature. For instance, Fisher et al. (2001) and Vlachos and Dekker (2003) assumed returns are a fixed proportion of sales. More recent studies, like Liang et al. (2020), proposed heuristic policies considering positive lead times and general return time distributions. Liang et al. (2023) offered a deterministic approximation for joint inventory control with demand substitution and product returns. Miao (2023) proposed an age-based accounting scheme for returns and an approximation algorithm for lost sales inventory systems. Bu et al. (2023) extended in this stream by studying a similar problem with lead times, showing that base-stock and myopic policies are asymptotically optimal with infinite lost sales penalties.

Regarding pricing decisions, the literature is notably less extensive and often considers returns with simplifying assumptions. Chen and Bell (2009) considered pricing when fixed proportions of sales are being returned. Hu et al. (2019) modeled pricing with the option of offering money-back guarantees.

A particularly relevant paper is Alptekinoğlu and Grasas (2014), which studies how consumer returns affect the structure of the optimal retail assortment in a static setting with known primitives. Their analysis shows that, under certain return-policy regimes, returns can fundamentally alter the structure of the optimal assortment, for example by making it desirable to carry both highly popular and more eccentric products. Our paper differs in two important ways. First, we study a dynamic learning problem in which purchase behavior, return behavior, and return timing are initially unknown and must be learned online. Second, our focus is not on characterizing the static structure of the optimal assortment under a particular return model, but on developing a parsimonious purchase–return model that is suitable for online learning under delayed and window-censored feedback. Thus, Alptekinoğlu and Grasas (2014) is complementary to our work: their paper provides a structural analysis of return-aware assortment design in a static setting, whereas ours develops a return-aware online learning framework with delayed and window-censored feedback. In a different dynamic setting, Rusmevichientong et al. (2020) study reusable products that are returned after use, while Wagner and Martínez-de Albéniz (2020) model customer search behavior as a dynamic program, incorporating the return process as customers sequentially try products and exchange until satisfied.

Most of the product-returns literature takes demand and return processes as given and studies how to incorporate returns into downstream operational decisions. We differ from this literature in two main ways.

First, purchase behavior, return behavior, and return timing are all initially unknown and must be learned dynamically from data. Second, return observations are delayed and censored by the return window, which makes the learning problem qualitatively different from standard dynamic assortment models without returns.

Bandits with delayed feedback. Our work is also related to bandits (and, more broadly, reinforcement learning) with delayed feedback. Early works on this stream such as Eick (1988) explored modeling the survival time of patients, incorporating delays into the decision-making process. In recent years, interest in this area has surged, leading to numerous significant contributions. For instance, Joulani et al. (2013) studied online learning scenarios where feedback has stochastic delayed feedback. Similarly, Chapelle (2014) examined delayed conversions in the context of display advertising, shedding light on the impact of delays on decision strategies. Further, Vernade et al. (2017) studied delays with known distributions specifically in the context of delayed conversions, providing insights into optimal decision-making for advertising.

Following these significant contributions, Pike-Burke et al. (2018) explored bandits with delayed, aggregated, and anonymous feedback. Meanwhile, Zhou et al. (2019) extended this exploration to generalized linear contextual bandits with delayed feedback. In a related vein, Vernade et al. (2020) investigated linear bandits with stochastic thresholded delays. Further adding to the discourse, Gael et al. (2020) tackled multi-armed bandits with arm-dependent stochastic delays, highlighting the complexity that arises when delays are not uniform across choices. Moreover, Cesa-Bianchi et al. (2022) considered composite and anonymous delayed feedback within non-stochastic bandits.

Our contribution to this literature is to study a problem with both immediate purchase feedback and delayed, window-censored return feedback, where the delayed feedback is structurally linked to the purchase process through shared latent primitives. The return-window mechanism introduces a distinctive information tradeoff: a longer window makes more returns observable, but also keeps more recent purchases unresolved for longer.

3. Model Description and Formulations

In this section, we develop a formal model for dynamic assortment optimization with product returns. We introduce a utility-based framework in which customer purchase and return behaviors are jointly determined by two latent utility primitives, capturing the key tradeoff between short-term product appeal and post-purchase satisfaction. We then describe how these primitives influence both the initial choice and eventual return decisions, and formulate the joint optimization and learning problem faced by the retailer.

Model framework. We consider a retailer offering a selection of N products over a finite selling season of T periods. Customers arrive sequentially and are shown an assortment $S_t \subseteq [N]$ at each period t. Based on a noisy utility perception, the customer selects either one product $i \in S_t$ or opts for no purchase. After a product is purchased, the customer subsequently decides whether to keep or return the product based on additional post-purchase utility signals. Returns may occur with delay and are only observable within a finite return window.

For each product i, we model two fundamental latent characteristics that drive customer behavior:

- q_i: Represents the customer's post-purchase satisfaction from keeping and using the product. This captures intrinsic product qualities such as fit, quality, and functionality. Products with high q_i generally satisfy customers and are unlikely to be returned.
- ρ_i: Represents the effective strictness or hassle associated with returning product i. It encompasses both objective factors (such as restocking policies, packaging requirements, or shipping frictions) and customers' subjective perception of how costly or inconvenient a return would be. Lower values of ρ_i correspond to more return-friendly products (i.e., higher propensity to return), while higher values of ρ_i make returns less attractive and therefore less likely.

The customer's decision process incorporates three sources of randomness that reflect real-world uncertainties:

- $\varepsilon_i \sim \text{Gumbel}(0, 1)$: Pre-purchase uncertainty about product match. Customers cannot perfectly assess q_i and ρ_i before purchase.
- $\zeta_{i,k} \sim \text{Gumbel}(0, 1)$: Post-purchase experience shock, capturing the difference between the customer's expected and actual satisfaction based on q_i.
- $\zeta_{i,r} \sim \text{Gumbel}(0, 1)$: Return execution shock. Random factors affecting the actual hassle of returning (mood, time constraints, drop-off proximity).

We assume that all shocks $\{\varepsilon_i\}$, $\{\zeta_{i,k}\}$, and $\{\zeta_{i,r}\}$ are mutually independent across products and periods.

Purchase Decision. The customer forms an expected utility for each product i at the time of purchase:

$$U_i = q_i - \rho_i + \varepsilon_i,$$

where the term ρ_i captures how return hassles reduce purchase appeal by increasing perceived risk. The customer selects the product with the highest utility among those in S_t, or opts for the no-purchase alternative (normalized to utility 0). The resulting choice probabilities follow the multinomial logit (MNL) model:

$$\Pr(a_t = i \mid S_t) = \frac{e^{q_i - \rho_i}}{1 + \sum_{j \in S_t} e^{q_j - \rho_j}}.$$

Let $D_{i,t}(S_t)$ denote the binary indicator of whether item i is purchased by the customer shown assortment S_t at period t; that is, $D_{i,t}(S_t) = 1$ if $a_t = i$, and 0 otherwise.

Return Process. After purchase, the customer experiences the product and decides whether to keep or return it. This decision compares the realized satisfaction against the effective return threshold:

$$\text{Return product } i \text{ if } \quad q_i + \zeta_{i,k} < -\rho_i + \zeta_{i,r}.$$

The left side represents the actual satisfaction derived from the product, while the right side represents the net hassle of returning (negative because high hassle makes returning less attractive). Since the difference of two i.i.d. Gumbel$(0, 1)$ variables follows a standard logistic distribution, the return probability becomes

$$r_i = \mathbb{P}(\text{product } i \text{ is returned}) = \sigma\big(-(q_i + \rho_i)\big),$$

where $\sigma(x) = 1/(1+e^{-x})$ is the sigmoid function.

REMARK 1. Our model is parameterized by two economically meaningful primitives for each product i: q_i, which captures expected post-purchase satisfaction (e.g., fit, quality, or functionality), and ρ_i, which captures return hassle or friction (e.g., inconvenience, packaging burden, or return-process strictness). These primitives jointly govern both purchase and return behavior. Before purchase, they shape the product's attractiveness by capturing both expected satisfaction and the customer's ex ante concern about having to return the product; after purchase, they determine the propensity to keep versus return the product once post-purchase uncertainty is realized.

The primitive representation is also useful for interpretation and counterfactual analysis. For example, it allows us to study how demand and returns change when return policies become more lenient (a decrease in ρ_i) while product satisfaction is held fixed, or when product quality improves (an increase in q_i) while return friction is unchanged.

This specification captures the retailer's central trade-off: more lenient return conditions (lower ρ_i) can make a product more attractive to purchase, but may also increase its likelihood of being returned, whereas stricter return conditions have the opposite effect. As a result, the retailer must learn primitives that simultaneously shape both demand and returns.

Censoring and return windows. For each product i and purchase period s, let $X_{i,s,t} = \mathbb{I}\{\text{the unit of product } i \text{ bought at } s \text{ is returned at } t\}$. Conditional on a purchase of product i at time s, let d_s denote the return delay of that unit. Under the baseline model, d_s has CDF

$$\Psi(m) := \Pr(d_s \leq m), \qquad m \in \mathbb{N},$$

with $\Psi(0) = 0$ and $\lim_{m\to\infty} \Psi(m) = 1$. Define the implied pmf by finite differences $f_d(\epsilon) := \Psi(\epsilon) - \Psi(\epsilon - 1)$ for $\epsilon \in \mathbb{N}$. Then

$$\Pr\!\big(X_{i,s,s+\epsilon} = 1 \,\big|\, D_{i,s}(S_s) = 1\big) = r_i\, f_d(\epsilon) = r_i\big(\Psi(\epsilon) - \Psi(\epsilon-1)\big), \quad \epsilon \in \mathbb{N}.$$

Even though the customer's return process $\{X_{i,s,t}\}_{t>s}$ is defined for all $t > s$, the retailer observes this process only for a limited period of time due to a fixed return window policy prevalent in the retail industry. Different retailers enforce different return windows. To capture this variety and to assess the impact of the return window on the retailer's objective, we denote the length of the fixed return policy by μ, meaning that returns are allowed only within μ periods of the purchase. This means that the retailer observes the return process $X_{i,s,t}$ for a purchase made at time s only while $t - s \leq \mu$. Accordingly, the fixed return window creates two forms of partial feedback. First, returns that would occur after μ are never observed as admissible in-window returns. Second, for recent purchases with $t - s < \mu$, the retailer does not yet know whether a return will occur before the window closes, so these observations are right-censored at the decision time.

This type of fixed-window truncation is operationally relevant in practice; see Ertekin and Agrawal (2021) for empirical evidence on how customer behavior over the return-period window affects retail outcomes. We assume that when a return is processed, the retailer can match it to the original sale record, so that both the product identity i and purchase time s are observed. This is standard in online retail settings with order-level transaction and return records. Consequently, the observable (censored) return process is given by:

$$X^{\mu}_{i,s,t} = X_{i,s,t} \cdot \mathbb{I}(t - s \leq \mu).$$

This censoring and delay introduce two critical estimation challenges. First, for each product sold, there exists uncertainty regarding whether it will eventually be returned, persisting until either the customer explicitly returns it or the return window μ expires. Second, returns intentions beyond the return window μ are unobservable, complicating the accurate estimation of return probabilities. Both issues are explicitly incorporated into our modeling and learning algorithms, thereby enhancing the robustness and practical applicability of our approach.

The decision-making process and the optimization problem. Combining all the above components, we now formulate the retailer's optimization problem under capacity constraints, uncertain demand, and returns. For each time step $t \in [T]$:

- The retailer selects an assortment $S_t \subset [N]$ of at most K products.
- Customers select $a_t \in S_t \cup \{0\}$, and the retailer earns p_i if $a_t = i$.
- Returns are processed within window μ, and each returned unit of product i incurs an expected net cost c_i (including refund, handling, and any loss of margin net of salvage).

Define the time-t observation as $O_t := \Big(a_t,\ \{X^{\mu}_{i,s,t} : 1 \leq s < t,\ i \in [N]\}\Big)$. Let $\mathcal{F}_t := \sigma(O_1, \ldots, O_t)$ be the natural filtration. We assume the policy is non-anticipating, i.e., S_t is $\mathcal{F}_{t-1}$-measurable for all t.

Letting Π denote the set of admissible non-anticipating policies and $\theta = \{q_i, \rho_i; i = 1, \ldots, N\}$ denote the true model parameters, the expected profit under policy $\pi = (S^{\pi}_1, \ldots, S^{\pi}_T) \in \Pi$ is:

$$R(\pi) = \sum_{t=1}^{T} \mathbb{E}^{\pi}_{\theta,\Psi}\left[\sum_{i \in S^{\pi}_t}\left(p_i D_{i,t}(S^{\pi}_t) - c_i \sum_{\epsilon=1}^{\mu} X_{i,t,t+\epsilon}\, D_{i,t}(S^{\pi}_t)\right)\right]. \tag{A}$$

where demand and returns are generated via the joint purchase-return process described above. In (A), the first term $p_i D_{i,t}(S^{\pi}_t)$ represents the revenue earned from the sale of product i at time t, while the second term $c_i \sum_{\epsilon=1}^{\mu} X_{i,t,t+\epsilon} D_{i,t}(S^{\pi}_t)$ captures the expected return cost for product i sold at t and returned within the window μ. Also, note that even though the planning horizon ends in period T, the return process continues up to period $T + \mu$.

Learning and regret. In this paper, we address the optimization problem (A) when the latent utility parameters q_i, ρ_i and the return delay distribution $\Psi(\cdot)$ are unknown a priori, and must learn them dynamically by observing sales and returns. This yields a classic exploration–exploitation tradeoff, formalized by the notion of regret:

DEFINITION 1. For a policy π, the (expected) regret is defined as

$$\text{Regret}(\pi) = R(\pi^*(\theta)) - R(\pi)$$

where $\pi^*(\theta, \Psi)$ denotes the optimal policy of the clairvoyant, i.e.,

$$\pi^*(\theta, \Psi) := \arg\max_{\pi\in\Pi} \sum_{t=1}^{T} \mathbb{E}^{\pi}_{\theta,\Psi}\left[\sum_{i\in S_t^\pi}\Big(p_i - c_i\sum_{\epsilon=1}^{\mu} X_{i,t,t+\epsilon}\Big)D_{i,t}(S_t^\pi)\right].$$

The above definition of regret quantifies the difference between the optimal expected profit of the clairvoyant and the expected profit achieved by policy π. Therefore, the goal is to design an online policy π that minimizes regret.

Technical assumptions. We have the following technical assumption which is common in the assortment optimization literature (e.g. Agrawal et al. (2019), Chen et al. (2023)).

ASSUMPTION 1. *There exists a constant $\bar{u} > 0$ such that for all $i \in [N]$, $-\bar{u} \le q_i + \rho_i \le \bar{u}$ and $-\bar{u} \le q_i - \rho_i \le \bar{u}$. Prices and return costs are scaled as $0 \le p_i \le 1$ and $0 \le c_i \le 1$.*

Assumption 1 says that the utility parameters lie in a compact set. Additionally, scaling prices and costs to be less than or equal to 1 is a standard approach in the literature, simplifying the analysis without loss of generality.

We further introduce our assumption regarding the return probabilities and the return delay distribution in the following statement.

ASSUMPTION 2. *Let d_s denote the (random) return delay, in periods, for a unit purchased at time s, with pmf $f_d(\cdot)$ on the positive integers. The return delay distribution f_d can be unbounded, but has finite expected value $\mathbb{E}[d_s]$, and the retailer knows an upper bound $\overline{d}$ on the expected return delay such that $\mathbb{E}[d_s] \le \overline{d}$.*

As we discuss later in Lemma EC.4, we can remove the assumption on return delay distribution without loss of generality when dealing with short return windows or return delay distributions with exponentially decaying tails. Indeed, observations from our dataset indicate that return delays predominantly follow exponentially decaying tail characteristics, which supports this assumption. Furthermore, much of the existing literature on product returns, refer to Abdulla et al. (2019), has assumed either deterministic or survival-type distribution for return delay, both of which also support the mildness of Assumption 2.

4. Proposed Algorithm for Return-Aware Assortments (COBRA)

We now develop **COBRA** (*Confidence Bounds for Return-Aware Assortments*), an epoch-based learning policy for dynamic assortment optimization with delayed and window-censored returns. The central difficulty is that the retailer must learn, simultaneously, the latent primitives governing purchase and return behavior and the return-delay profile, while continuing to make assortment decisions. Because return outcomes are revealed only gradually, recent purchases generate censored feedback at the time the next assortment decision is made.

COBRA addresses this challenge through three components. First, it uses an epoch structure that converts repeated assortment offers into tractable per-product purchase statistics. Second, it updates product-level parameters through a penalized likelihood that combines purchase observations with delayed and censored return data, together with a separate estimate of the return-delay distribution. Third, it selects assortments optimistically over confidence sets for the latent primitives and the observable in-window return mass. Figure 1 provides a schematic overview of this learning-and-optimization cycle.

4.1. Epoch structure and sufficient statistics

COBRA partitions time into epochs separated by no-purchase events. Let $t_0 := 0$, and for each $\tau \geq 1$, let t_τ denote the first time after $t_{\tau-1}$ at which a no-purchase occurs while a fixed assortment is being repeated. In epoch τ, the algorithm selects an assortment S_τ and offers it repeatedly over periods $t = t_{\tau-1}+1, \ldots, t_\tau$ until the first no-purchase occurs.

For each product i, let

$$\mathcal{A}_{i,\tau} := \{t \leq t_\tau :\ i \in S_t\}, \qquad n_{i,\tau} := |\mathcal{A}_{i,\tau}|,$$

denote the exposure set and cumulative exposure count up to the end of epoch τ, and let

$$\mathcal{B}_{i,\tau} := \{s \leq t_\tau :\ a_s = i\}$$

be the set of purchase times of product i. We further distinguish purchases whose return windows have already closed by time t_τ:

$$\mathcal{B}^{\text{full}}_{i,\tau} := \{s \in \mathcal{B}_{i,\tau} : s \leq t_\tau - \mu\}, \qquad P^{\text{full}}_\tau := \{(i,s) : i \in [N],\ s \in \mathcal{B}^{\text{full}}_{i,\tau}\}.$$

For each matured purchase $(i,s) \in P^{\text{full}}_\tau$, define the in-window return indicator

$$Z_{i,s} := \sum_{\epsilon=1}^{\mu} X^{\mu}_{i,s,s+\epsilon} \in \{0,1\}.$$

Under the model,

$$\Pr(Z_{i,s} = 1 \mid \theta, \Psi) = \psi_\mu\, \sigma\big(-(q_i + \rho_i)\big), \qquad \psi_\mu := \Psi(\mu).$$

The main advantage of the epoch construction is that, although customer choices follow an MNL model, repeating a fixed assortment until the first no-purchase induces a per-product geometric structure for purchase counts. This decoupling property is standard in the MNL bandit literature (see, e.g., Agrawal et al. (2019)) and allows the purchase component of the likelihood to be written in per-product form despite the original MNL coupling.

4.2. Per-product likelihood and penalized updates

Fix a product i and write $\theta_i = (q_i, \rho_i)$. Define

$$\alpha_i(\theta_i) := \sigma\big(-(q_i - \rho_i)\big) = \frac{1}{1 + \exp(q_i - \rho_i)}, \qquad r_i(\theta_i) := \sigma\big(-(q_i + \rho_i)\big).$$

Here $\alpha_i(\theta_i)$ is the geometric parameter induced by the epoch structure, whereas $r_i(\theta_i)$ is the return probability conditional on purchase.

Let $L_{i,\tau} := \big|\{\ell \le \tau :\ i \in S_\ell\}\big|$ be the number of epochs up to t_τ in which product i is offered, and for each such epoch ℓ, let $N_{i,\ell}$ be the number of purchases of i before the stopping no-purchase. For each purchase time $s \in \mathcal{B}_{i,\tau}$, define the observed return-window length

$$m_{s,\tau} := \min\{\mu,\ t_\tau - s\} \in \{0, 1, \dots, \mu\}.$$

Then the censored per-product log-likelihood up to epoch τ is

$$\begin{aligned} \ell_{i,\tau}(\theta_i, \Psi) = &\sum_{\ell=1}^{L_{i,\tau}} \Big[N_{i,\ell} \log(1 - \alpha_i(\theta_i)) + \log \alpha_i(\theta_i) \Big] \\ &+ \sum_{s \in \mathcal{B}_{i,\tau}} \left\{ \sum_{\epsilon=1}^{m_{s,\tau}} X^{\mu}_{i,s,s+\epsilon} \big[\log r_i(\theta_i) + \log f_d(\epsilon) \big] + \Big(1 - \sum_{\epsilon=1}^{m_{s,\tau}} X^{\mu}_{i,s,s+\epsilon} \Big) \log\Big(1 - r_i(\theta_i) \Psi(m_{s,\tau}) \Big) \right\}. \end{aligned} \tag{1}$$

The first line corresponds to purchase observations induced by the epoch structure, while the second line incorporates return information subject to delay and right-censoring.

We also introduce the corresponding oracle likelihood in which every purchase is observed through the full return window of length μ. This oracle object is not available to the algorithm, but serves as an analytical benchmark: the gap between the observed and oracle likelihoods isolates the effect of unresolved recent purchases and is analyzed in Appendix EC.1.4.

At the beginning of epoch $\tau + 1$, COBRA updates product i by solving the penalized maximum likelihood problem

$$\hat{\theta}_{i,\tau} \in \arg\max_{\theta_i \in \Theta_i} \Big\{ \ell_{i,\tau}(\theta_i, \widehat{\Psi}^\tau) - \tfrac{\eta}{2} \|\theta_i\|_2^2 \Big\}, \tag{2}$$

where Θ_i is the compact parameter set and $\widehat{\Psi}^\tau$ is the current estimate of the delay profile. The ridge term stabilizes estimation in early epochs when return information is still sparse.

The main technical issue is that (1) is not the full-window likelihood: recent purchases have only partially revealed return outcomes by epoch τ. To handle this, we compare the observed censored likelihood with an oracle likelihood in which every purchase is observed through the full return window. This oracle–observed comparison isolates the effect of unresolved recent purchases and is the key step through which the return window enters the regret analysis.

4.3. Estimating the delay distribution

To evaluate the censored likelihood in (1), COBRA estimates the return-delay CDF

$$\Psi(m) = \Pr(d \le m), \qquad m = 1, \dots, \mu,$$

using matured purchases only. For each matured unit $(i,s) \in P_\tau^{\text{full}}$, define the cumulative return indicator

$$Z_{i,s}^{\le m} := \sum_{\epsilon=1}^{m} X_{i,s,s+\epsilon}^{\mu} \in \{0,1\}.$$

Since

$$\mathbb{E}[Z_{i,s}^{\le m} \mid \theta] = \Psi(m)\, r_i(\theta_i),$$

we normalize by the aggregate predicted return mass

$$S_\tau^{\text{full}} := \sum_{(i,s)\in P_\tau^{\text{full}}} r_i(\hat{\theta}_{i,\tau-1}),$$

and estimate the delay profile by

$$\widehat{\Psi}^\tau(m) := \left[\frac{\sum_{(i,s)\in P_\tau^{\text{full}}} Z_{i,s}^{\le m}}{S_\tau^{\text{full}}} \right]_{[0,1]}, \qquad m = 1, \dots, \mu.$$

We write $\widehat{\psi}_\mu^\tau := \widehat{\Psi}^\tau(\mu)$ for the estimated in-window return mass.

For the optimistic step, only a lower confidence bound on ψ_μ is required, because the profit objective is monotone in this scalar. Accordingly, COBRA uses

$$\underline{\psi}_\tau := \max\{0, \widehat{\psi}_\mu^\tau - \beta_\tau\},$$

where β_τ is a concentration radius derived in Appendix A.5. The full profile $\widehat{\Psi}^\tau$ is used in the likelihood update, while the scalar bound $\underline{\psi}_\tau$ is sufficient for optimistic assortment selection.

Two features of this estimator are important for the analysis. First, the estimator uses matured purchases only, which avoids direct censoring bias from unresolved recent purchases. Second, the normalization uses the current plug-in return-probability estimates, so the resulting error bound contains both an empirical fluctuation term and a denominator-mismatch term. These two effects are controlled in Appendix A.5.

4.4. Confidence sets and optimistic assortment selection

Given the product-level estimates $\hat{\theta}_{i,\tau}$, COBRA constructs a confidence box

$$C_{i,\tau} := \{\theta_i : \|\theta_i - \hat{\theta}_{i,\tau}\|_\infty \le W_{i,\tau}\}, \qquad C_\tau := \bigotimes_{i=1}^{N} C_{i,\tau},$$

where the radius $W_{i,\tau}$ depends on the exposure count $n_{i,\tau}$, the return-window length μ, and the delay-estimation error β_τ. The exact form of the radius is given in Lemma EC.7.

For any feasible assortment S with $|S| \le K$, define the expected per-customer profit

$$Q(S;\theta,\psi) := \sum_{i\in S}\big(p_i - c_i\psi\, r_i(\theta_i)\big)\frac{\exp(q_i-\rho_i)}{1+\sum_{j\in S}\exp(q_j-\rho_j)}. \tag{3}$$

At the start of epoch τ, COBRA selects an assortment by optimism:

$$S_\tau \in \arg\max_{S:|S|\le K}\left\{\max_{\theta\in C_\tau,\ \psi\in[\underline{\psi}_\tau,1]} Q(S;\theta,\psi)\right\}.$$

Thus, COBRA chooses the assortment with the highest plausible return-aware profit under the current uncertainty. As the confidence sets shrink over time, the policy progressively concentrates on assortments that are close to optimal under the true model parameters. Algorithm 1 summarizes the resulting procedure.

The technical analysis of COBRA is developed in Appendix A. The proof proceeds in four steps. First, the epoch structure converts the purchase component into per-product geometric observations, yielding tractable product-level likelihoods. Second, we establish regularity and concentration properties of the penalized censored likelihood, together with a bias bound that quantifies the effect of unresolved recent purchases and delay-profile estimation error. Third, these results imply valid confidence radii for both the product-level primitives and the in-window return mass. Finally, a UCB-style epoch decomposition yields the regret bound.

THEOREM 1 **(Regret of COBRA).** *Under Assumptions 1 and 2, Algorithm 1 satisfies*

$$\mathrm{Regret}(\pi_{\mathrm{COBRA}}) \le \frac{C}{\psi_\mu}\left(\sqrt{NT\,\ln(NT)} + N\ln T\cdot\min\left\{\mu,\ \sqrt{\mu\ln(NT)}+\overline{d}\right\}\right) + \frac{C}{\psi_\mu^2}\left(\sqrt{T\ln(NT\mu)}+\ln T\right),$$

for a universal constant $C>0$ depending only on $\bar{u}$, K, η, and absolute numerical constants.

The theorem highlights two distinct effects of delayed returns. The factor ψ_μ^{-1} captures the statistical difficulty of learning from returns when only a small fraction of eventual returns are observed within the return window. The term

$$\min\left\{\mu,\ \sqrt{\mu\ln(NT)}+\overline{d}\right\}$$

captures the effect of censoring from unresolved recent purchases: in the worst case this dependence is linear in the return window, whereas under lighter-tailed delays it improves to a sublinear dependence on μ.

Proof sketch. The proof combines four ingredients. First, the epoch structure yields per-product geometric purchase statistics, which make the purchase component of the likelihood tractable. Second, we establish concentration for the oracle score process and control the discrepancy between the oracle and observed likelihoods through a censoring-bias decomposition. Third, we use these ingredients together with curvature bounds for the penalized likelihood to derive high-probability confidence radii for the product-level primitives and the delay profile. Finally, on the event that these confidence bounds hold uniformly over epochs, optimism implies that the per-epoch regret is controlled by the same confidence radii, and summing over epochs gives the stated bound. The full proof is given in Appendix EC.1.8.

At a technical level, the analysis hinges on an oracle–observed likelihood comparison for censored returns and on a uniform error bound for the delay-profile estimator based on matured purchases.

Algorithm 1: Confidence Bounds for Return-Aware Assortment (COBRA)

Input. Horizon T, return window μ, delay bound $\overline{d}$, capacity K, confidence level δ, regularization η.

Initialize. $t \leftarrow 1$, $\tau \leftarrow 0$;

initialize $\hat{\theta}_{i,0} = (\hat{q}_{i,0}, \hat{\rho}_{i,0})$ for all $i \in [N]$;

initialize $\widehat{\Psi}^0$ and $\underline{\psi}_0$.

while $t \le T$ **do**

  $\tau \leftarrow \tau + 1$;

  Using matured purchases and $\{\hat{\theta}_{i,\tau-1}\}_{i=1}^N$, update the delay-profile estimate $\widehat{\Psi}^\tau$; compute β_τ as in Lemma EC.5; set $\widehat{\psi}_\mu^\tau := \widehat{\Psi}^\tau(\mu)$, $\underline{\psi}_\tau := \max\{0, \widehat{\psi}_\mu^\tau - \beta_\tau\}$.

  For each product $i \in [N]$, update

$$\hat{\theta}_{i,\tau} \in \arg\max_{\theta_i \in \Theta_i} \left\{ \ell_{i,\tau}(\theta_i, \widehat{\Psi}^\tau) - \frac{\eta}{2} \|\theta_i\|_2^2 \right\},$$

  and construct

$$C_{i,\tau} := \{\theta_i : \|\theta_i - \hat{\theta}_{i,\tau}\|_\infty \le W_{i,\tau}\}, \qquad C_\tau := \bigotimes_{i=1}^N C_{i,\tau},$$

  where $W_{i,\tau}$ is the radius from Lemma EC.7.

  Choose

$$S_\tau \in \arg\max_{S:|S| \le K} \left\{ \max_{\theta \in C_\tau} Q(S; \theta, \underline{\psi}_\tau) \right\}.$$

  repeat

    Offer S_τ at time t;

    observe $a_t \in S_\tau \cup \{0\}$;

    observe all returns arriving at time t and update statistics;

    $t \leftarrow t + 1$.

  until $a_t = 0$ or $t > T$;

end

5. Efficient Solution of the Optimistic Assortment Problem

A key step of COBRA is the optimistic assortment selection at epoch τ:

$$S_\tau \in \arg\max_{S:|S| \le K} \left\{ \max_{\theta \in C_\tau} Q(S; \theta, \underline{\psi}_\tau) \right\}, \tag{4}$$

where $C_\tau = \bigotimes_{i=1}^N C_{i,\tau}$ is the product confidence set and $\underline{\psi}_\tau$ is the lower confidence bound on the in-window return mass. An illustration of this optimistic parameter selection within a product's confidence set is given in Figure 2. A brute-force search over all $\binom{N}{K}$ feasible assortments is computationally intractable. In this section, we show that (4) can be solved efficiently by adapting the reformulation technique of Rusmevichientong et al. (2010) to our return-aware setting.

Algorithm 2: Oracle for the Optimistic Assortment Problem

```
Input: Confidence sets C_{i,τ} for all i ∈ [N], lower confidence bound ψ̲_τ, capacity K.
Output: Optimistic assortment S_τ.
Set search range [λ_low, λ_high] ← [0, max_i p_i].
while λ_high − λ_low > ε do
    λ_mid ← (λ_low + λ_high)/2.
    for each product i ∈ [N] do
        Compute h*_{i,τ}(λ_mid) by solving (5).
    end
    Let P* be the set of the K largest positive values among {h*_{1,τ}(λ_mid), ..., h*_{N,τ}(λ_mid)}.
    Set g*_τ(λ_mid) ← Σ_{h∈P*} h.
    if g*_τ(λ_mid) ≥ λ_mid then
        λ_low ← λ_mid.
    else
        λ_high ← λ_mid.
    end
end
λ* ← λ_low.
for each product i ∈ [N] do
    Compute h*_{i,τ}(λ*).
end
Return the set of K products with the largest positive values among {h*_{i,τ}(λ*)}_{i=1}^N.
```

For a fixed epoch τ, define the optimistic value of an assortment S by

$$Q_\tau^*(S) := \max_{\theta \in C_\tau} Q(S; \theta, \underline{\psi}_\tau),$$

and let

$$Z_\tau^* := \max_{S:|S| \le K} Q_\tau^*(S).$$

For any threshold $\lambda \ge 0$, the condition $Q_\tau^*(S) \ge \lambda$ can be rewritten in a form that separates across products. This leads to the product-level optimistic potential

$$h_{i,\tau}^*(\lambda) := \max_{\theta_i \in C_{i,\tau}} \left[e^{q_i - \rho_i} \left(p_i - c_i \underline{\psi}_\tau \, \sigma(-(q_i + \rho_i)) - \lambda \right) \right]. \tag{5}$$

For a fixed λ, the maximum left-hand side is obtained by selecting the K largest positive values among $\{h_{i,\tau}^*(\lambda)\}_{i=1}^N$. Let $g_\tau^*(\lambda)$ denote the sum of these K largest positive values. Then the optimistic assortment problem reduces to finding the largest λ such that

$$g_\tau^*(\lambda) \ge \lambda.$$

This reformulation yields an efficient oracle: perform a binary search over λ, compute $h^*_{i,\tau}(\lambda)$ for each product by low-dimensional optimization over $C_{i,\tau}$, and at the end select the K products with the largest positive potentials. Algorithm 2 summarizes the procedure. The correctness of the reformulation and the monotonicity of $g^*_\tau(\lambda) - \lambda$ are given in Appendix EC.2. As a result, the optimistic assortment can be computed to arbitrary numerical precision in time polynomial in N.

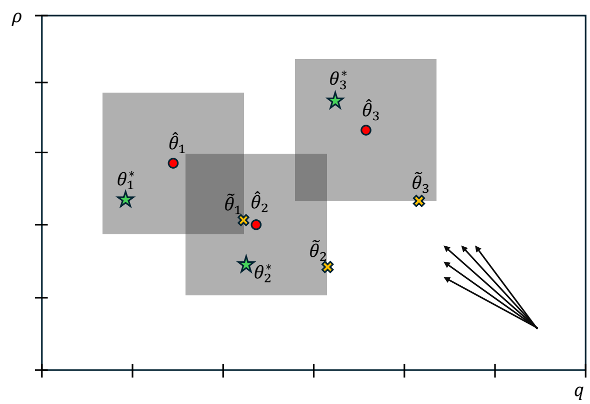

Figure 2 **Illustration of the optimistic parameter choice for a product in primitive space. The rectangle represents the confidence set $C_{i,\tau}$ for (q_i, ρ_i). The point θ^*_i denotes the true parameter, and $\hat{\theta}_{i,\tau}$ denotes the maximum likelihood estimate. The rays schematically indicate directions of increasing optimistic objective within the confidence set. The optimistic point $\tilde{\theta}_i = (\tilde{q}_i, \tilde{\rho}_i)$ maximizes the product's contribution to the return-aware objective and need not coincide with a coordinate-wise extremum.**

6. A Regret Lower Bound

We now show that the leading $\sqrt{NT}$ dependence in Theorem 1 is unavoidable, and that delayed in-window return observability enters the lower bound as well. Our proof of the lower bound theorem works on a carefully constructed scenario where we set certain parameters to establish the minimal achievable performance.

Fix a value $\psi_\mu \in (0, 1]$ and set $p_i = c_i = 1$ for all $i \in [N]$. Consider two disjoint subsets

$$S^{\text{att}}, S^{\text{ret}} \subseteq [N], \qquad |S^{\text{att}}| = |S^{\text{ret}}| = K/2,$$

with $K \le N/4$. The set S^{att} contains products that are slightly better along the *purchase-attractiveness* index $q_i - \rho_i$, while S^{ret} contains products that are slightly better along the *return* index $-(q_i + \rho_i)$. Specifically, for parameters $\epsilon_1 \in (0, 1/2]$ and $\epsilon_2 \in (0, 1/4]$, define the hard instance through

$$e^{q_i - \rho_i} = \begin{cases} \frac{2(1+\epsilon_1)}{K}, & i \in S^{\text{att}}, \\ \frac{2}{K}, & i \notin S^{\text{att}}, \end{cases} \qquad \sigma\big(-(q_i + \rho_i)\big) = \begin{cases} \frac{1}{2} - \epsilon_2, & i \in S^{\text{ret}}, \\ \frac{1}{2}, & i \notin S^{\text{ret}}. \end{cases} \tag{6}$$

Because the map

$$(q_i, \rho_i) \mapsto \big(q_i - \rho_i, -(q_i + \rho_i)\big)$$

is invertible, (6) defines a valid choice of primitives (q_i, ρ_i). The optimal size-K assortment for this instance is

$$S^\star := S^{\text{att}} \cup S^{\text{ret}}.$$

The lower bound is stated next.

THEOREM 2. *Suppose $K \le N/4$. Then for any non-anticipating policy π, there exists an instance of the return-aware assortment learning problem with in-window return mass ψ_μ such that*

$$\text{Regret}(\pi) \ \ge\ \Omega\left(\left(1 - \frac{\psi_\mu}{2}\right)\sqrt{NT} + \sqrt{\frac{NT\psi_\mu}{K}}\right).$$

The lower bound has two components. The first term comes from having to learn which products are slightly better in the purchase-attractiveness index $q_i - \rho_i$, while the second term comes from having to learn which products are slightly better in the return index $-(q_i + \rho_i)$ using only in-window return observations. When ψ_μ is treated as a fixed constant, the theorem recovers the familiar $\Omega(\sqrt{NT})$ rate from dynamic assortment learning without returns.

Proof idea. We sketch the main steps here and defer the full proof to Appendix EC.4.

Step 1: Quantify the instantaneous regret by misses in the two critical subsets. For any size-K assortment S, define

$$\delta_{\text{att}} := 1 - \frac{2|S \cap S^{\text{att}}|}{K}, \qquad \delta_{\text{ret}} := 1 - \frac{2|S \cap S^{\text{ret}}|}{K}.$$

Thus δ_{att} and δ_{ret} measure how far S is from containing all products from the attraction-critical set and the return-critical set, respectively. A direct calculation under (6) shows that

$$Q(S^\star) - Q(S) \ \ge\ \frac{\delta_{\text{att}}\epsilon_1\left(1 - \frac{\psi_\mu}{2} - \epsilon_2\psi_\mu\right) + \delta_{\text{ret}}\epsilon_2\psi_\mu(3 + \epsilon_1)}{16}. \tag{7}$$

Hence cumulative regret is lower bounded by the expected cumulative number of misses from S^{att} and S^{ret}.

Step 2: Average over a hard family of instances. We next average over all possible choices of S^{att} with S^{ret} fixed, and symmetrically over all possible choices of S^{ret} with S^{att} fixed. This reduces the problem to bounding how many times the policy can correctly identify products that belong to one of the two critical subsets.

Here and below, $a \lesssim b$ means that $a \le Cb$ for some positive universal constant C, and $a \asymp b$ means that $a \lesssim b$ and $b \lesssim a$.

Step 3: Control the purchase part via a neighboring-instance KL argument. For the S^{att} component, we adapt the information-theoretic lower-bound argument of Chen and Wang (2018) for capacitated MNL-bandits. In particular, we compare two neighboring environments that differ only in whether a single product

i belongs to the attraction-critical set. Let P and G denote the induced laws over the full observation history. If $\widetilde{N}_i$ denotes the number of times product i is offered, then

$$\left|\mathbb{E}_P[\widetilde{N}_i] - \mathbb{E}_G[\widetilde{N}_i]\right| \le T\,\mathrm{TV}(P,G) \le T\sqrt{\tfrac{1}{2}D_{\mathrm{KL}}(P\|G)}, \tag{8}$$

where the second inequality is Pinsker's inequality. The only difference between the two environments is the purchase distribution when product i is offered, and the one-step KL divergence is of order $O(\epsilon_1^2/K)$. Summing over time via the chain rule implies

$$D_{\mathrm{KL}}(P\|G) \lesssim \frac{\epsilon_1^2}{K}\,\mathbb{E}_P[\widetilde{N}_i].$$

Averaging this bound over the hard family and choosing

$$\epsilon_1 \asymp \sqrt{\frac{N}{T}}$$

yields the contribution

$$\Omega\left(\left(1-\frac{\psi_\mu}{2}\right)\sqrt{NT}\right).$$

Step 4: Control the return part via KL and a χ^2 bound. For the S^{ret} component, consider two neighboring environments that differ only in whether product i belongs to the return-critical set. In this case the purchase probabilities are identical; the two environments differ only through the return observation attached to product i. Conditional on offering and purchasing product i, the in-window return signal is Bernoulli with means

$$\psi_\mu/2 \qquad \text{and} \qquad \psi_\mu(1/2-\epsilon_2).$$

Thus the information distinguishing the two environments is weaker and is scaled by ψ_μ. To bound the resulting divergence, we upper bound the Bernoulli KL divergence by the corresponding χ^2 distance, which gives a one-step bound of order

$$D_{\mathrm{KL}} \lesssim \psi_\mu \epsilon_2^2.$$

Using again (8), together with the chain rule over time and averaging over the hard family, we obtain the return-learning contribution

$$\Omega\left(\sqrt{\frac{NT\psi_\mu}{K}}\right)$$

by choosing

$$\epsilon_2 \asymp \sqrt{\frac{N}{TK\psi_\mu}}.$$

Combining the two contributions proves Theorem 2.

REMARK 2. Theorem 2 recovers the familiar $\Omega(\sqrt{NT})$ minimax scaling when ψ_μ is treated as a fixed constant. More importantly, together with the upper bound in Theorem 1, it shows that the effect of the return window is inherently two-sided rather than a proof artifact. As the window expands, a larger fraction $\psi_\mu = \Psi(\mu)$ of eventual returns becomes observable in-window, which makes return-sensitive products statistically easier to distinguish. At the same time, a longer window also means that more recent purchases remain unresolved for longer, so the learner must act under a heavier burden of delayed and censored feedback. Thus, the return window creates a fundamental information–delay trade-off: increasing μ improves return observability, but it also intensifies the temporal difficulty of learning from returns. The fact that this same qualitative tension appears in both the upper and lower bounds indicates that it is intrinsic to the problem.

7. Numerical Study

This section evaluates the performance of COBRA on both synthetic instances and a real dataset from our retail collaborator. The experiments have three objectives. First, they quantify the value of incorporating return behavior into dynamic assortment learning. Second, they compare COBRA with natural benchmark policies under delayed and window-censored return feedback. Third, they evaluate the practical performance of the algorithm on real transaction and return data.

To facilitate comparisons across instances, we report the *relative regret* of a policy π:

$$\mathrm{RR}(\pi) := \frac{R(\pi^*(\theta,\Psi)) - R(\pi)}{R(\pi^*(\theta,\Psi))},$$

where $\pi^*(\theta,\Psi)$ denotes the clairvoyant policy that knows the true primitives and the return-delay profile. In addition to profit-based performance, we also report return-volume outcomes separately when relevant, since return volume is an important operational metric even though the formal objective of the algorithm is profit maximization.

Benchmark policies. We compare COBRA with two benchmarks.

Explore-then-commit (ETC). This policy first allocates an initial exploration phase to collect purchase and return data, and then commits to the assortment estimated to be best for the remainder of the horizon. Unlike COBRA, it does not continuously update assortments as delayed return information arrives. The detailed benchmark construction is given in Appendix EC.5.1.

MNL-Bandit. Our second benchmark is the MNL-Bandit algorithm of Agrawal et al. (2019), which learns purchase behavior but ignores returns in both estimation and assortment optimization. Comparing COBRA with this benchmark isolates the value of incorporating return-aware learning into the decision process.

Synthetic-data setup. Unless otherwise stated, the synthetic experiments use $N = 10$ products, capacity $K = 4$, unit prices $p_i = 1$, and return costs $c_i = 1.1$ for all $i \in [N]$. We generate product primitives (q_i, ρ_i) so that the synthetic instances exhibit a nontrivial tradeoff between purchase attractiveness and return behavior.

To control the difficulty of learning on the purchase side, we introduce a parameter $\epsilon_1 > 0$ that governs the separation between the purchase indices $q_i - \rho_i$ of the best products and their closest competitors. Smaller values of ϵ_1 make purchase behavior harder to learn because the relevant products become less distinguishable in the choice model. Similarly, we introduce $\epsilon_2 > 0$ to control the difficulty of learning return behavior by varying the separation in the return indices $q_i + \rho_i$.

In the baseline configuration, the primitives are chosen so that products $1, 2, 4, 9$, and 10 have slightly stronger purchase appeal than the remaining products, while product 4 has a higher return tendency than all other products. This creates a setting in which the retailer faces a genuine tradeoff: some products are attractive in the purchase stage but less desirable once return costs are taken into account. Under this construction, the optimal assortment is $\{1, 2, 9, 10\}$.

Return delays are generated from a geometric distribution with mean $3T/40$, and the return window is set to $3T/20$. For each synthetic instance, we run 20 independent replications and report averages together with standard errors.

7.1. Experiment 1: Value of Incorporating Returns

Impact on profit and return volume. In this experiment, we evaluate the value of incorporating return behavior into dynamic assortment learning. We compare COBRA against three ETC benchmarks with exploration lengths $T^{0.7}$, $T^{0.8}$, and $T^{0.9}$, as well as MNL-Bandit, which learns purchase behavior but ignores returns in both estimation and assortment optimization.

Figure 3 reports relative regret as the horizon length increases. COBRA consistently achieves the best profit performance among the return-aware policies and substantially outperforms MNL-Bandit. This shows that explicitly incorporating return information into the learning-and-optimization loop improves long-run decision quality relative to a benchmark that treats returns as irrelevant. The ETC benchmarks also benefit from using return information, but their fixed exploration phase makes them less adaptive than COBRA, especially as more delayed feedback becomes available over time.

Tables 1 and 2 report the percentage reduction in return volume relative to MNL-Bandit. Several patterns emerge. First, all return-aware policies reduce returns relative to MNL-Bandit, confirming that ignoring returns leads to assortments with systematically higher return volume. Second, the ETC benchmarks often achieve larger reductions in return volume than COBRA, particularly when the exploration phase is longer. This is not surprising: ETC commits after a large amount of exploration and can therefore be more conservative with respect to high-return products, but this comes at the cost of worse profit performance, as seen in Figure 3. Third, COBRA still delivers substantial return reductions, and these reductions become more

Table 1 **Reduction in returns compared to MNL Bandit.**

Horizon length:	100	500	1000	2000	4000	8000
ETC $T^{0.7}$	19.2%	11.4%	12.1%	11.8%	13.0%	12.5%
ETC $T^{0.8}$	19.0%	13.4%	17.2%	17.8%	22.4%	23.8%
ETC $T^{0.9}$	23.0%	20.9%	19.0%	21.4%	25.8%	24.2%
COBRA	12.2%	9.3%	7.7%	16.8%	22.2%	21.3%

Table 2 **Reduction in returns compared to MNL Bandit.**

Return window size:	Very Short	Short	Normal	Long	Very Long
ETC $T^{0.7}$	7.5%	10.9%	13.4%	15.4%	16.0%
ETC $T^{0.8}$	9.3%	18.1%	23.5%	24.7%	25.3%
ETC $T^{0.9}$	17.9%	23.0%	25.6%	25.6%	25.4%
COBRA	4.9%	11.9%	21.7%	22.4%	21.2%

pronounced as the horizon grows and as the return window becomes longer. In particular, for moderate to large horizons and for normal-to-long return windows, COBRA achieves around 17–22% fewer returns than MNL-Bandit.

Taken together, these results highlight the main tradeoff in this experiment. If the sole goal were to suppress return volume, then a heavily exploratory ETC policy can sometimes do better. However, COBRA is designed to optimize return-aware profit rather than return volume alone. The numerical results show that it achieves the best regret performance while still generating materially fewer returns than a benchmark that ignores returns entirely. This confirms that modeling returns explicitly can improve profitability and produce a meaningful side benefit in terms of lower return volume.

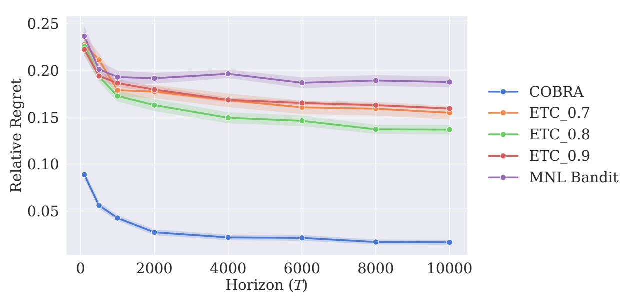

Figure 3 **Relative regret as a function of the horizon length. COBRA consistently achieves the lowest regret among the compared policies. The shaded regions represent standard errors.**

Impact of the return window. We next examine how the return-window length affects performance. The simulated return-window settings are Very Short (7 days), Short (15 days), Normal (30 days), Long (45 days), and Very Long (150 days).

Figure 4 reports the relative regret of the benchmark policies and of COBRA as the return window varies. Two patterns emerge. First, as shown in the left panel, the ETC benchmarks become less competitive as

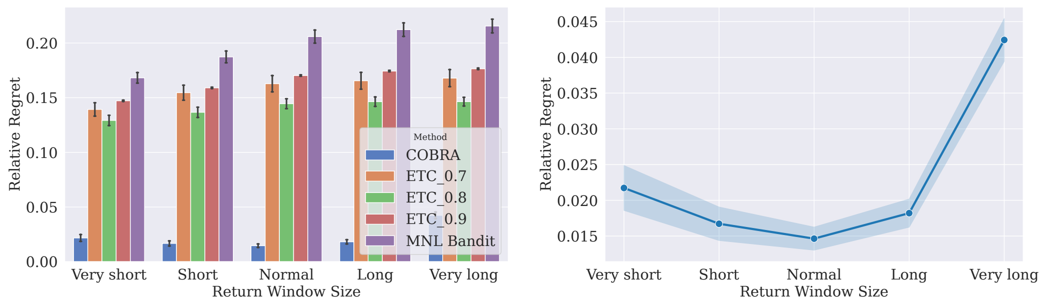

Figure 4 **Relative regret as a function of the return-window length. Left panel: all benchmark policies. Right panel: COBRA only. Error bars and shaded regions represent standard errors.**

the return window increases. This is natural: when the return window is short, return feedback is revealed relatively quickly, so a fixed exploration phase can still provide useful information before commitment. As the return window becomes longer, however, more return outcomes remain unresolved during exploration, which makes a one-shot commit decision less reliable.

Second, the right panel isolates COBRA and shows a non-monotone dependence on the return window. Relative regret is higher when the return window is very short, decreases for short-to-long windows, and then rises again for the very long window. This pattern is consistent with the tradeoff highlighted by our theoretical analysis. When the return window is very short, only a limited fraction of the return process is observed, so return-aware learning is less informative. When the window is moderately long, COBRA can better exploit return information and its performance improves. When the window becomes very long, delayed and unresolved returns create more severe censoring, which makes the learning problem harder again. Thus, the numerical results reflect the same tension as in the regret bound: return-aware learning becomes more valuable as the return window expands, but excessively long windows also increase the informational difficulty of the problem.

Impact of return delay. In this experiment, we vary the return-delay distribution while holding the in-window return mass fixed. Specifically, we consider three delay profiles(left-skewed, symmetric, and right-skewed) as illustrated in Figure 5. These cases allow us to examine how the timing of return information affects learning performance. In all three scenarios, we keep ψ_μ fixed, so that the total probability of observing a return within the window is the same across delay profiles. Thus, any performance differences are driven by the timing of the feedback rather than by changes in the overall volume of in-window returns.

The results are shown in the left panel of Figure 6. Relative regret is smallest under the left-skewed delay distribution and increases as the delay profile becomes more right-skewed. This pattern is intuitive: when returns arrive earlier, COBRA can incorporate the information more quickly into its assortment decisions, whereas right-skewed delays postpone informative feedback and make return-aware learning more difficult. The symmetric case lies between these two extremes. Overall, the experiment confirms that, even when

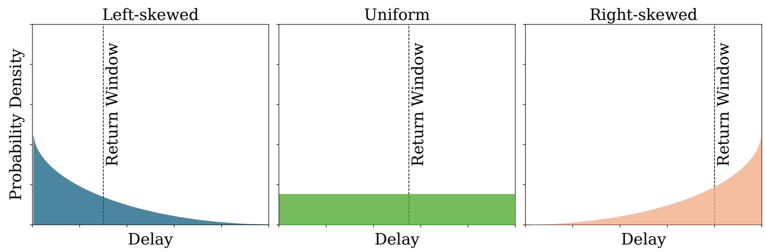

Figure 5 **Different patterns of return delay distribution.**

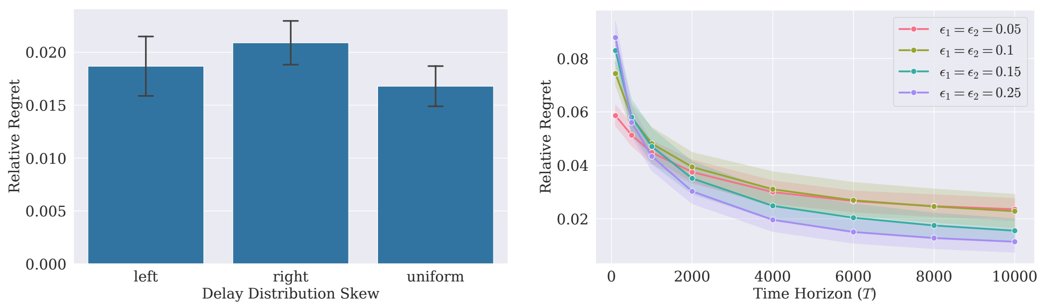

Figure 6 **Left panel: relative regret under different return-delay profiles. Right panel: relative regret as a function of the horizon length for different values of ϵ_1 and ϵ_2. Error bars and shaded regions represent standard errors.**

the total in-window return mass is held fixed, the timing of return feedback has a significant effect on performance.

Robustness with respect to ϵ_1 and ϵ_2. In the right panel of Figure 6, we vary $\epsilon_1 = \epsilon_2$ to study how the separation between the optimal assortment and its competitors affects learning performance. These parameters play two competing roles. Larger values of ϵ_1 and ϵ_2 make the instance easier to distinguish statistically, since the purchase and return primitives of the relevant products are more clearly separated. At the same time, larger separation also makes early mistakes more costly, because pulling a suboptimal assortment leads to a larger profit loss. Conversely, when ϵ_1 and ϵ_2 are small, the problem is harder to identify, but the cost of choosing a near-optimal assortment is also smaller.

This tradeoff is clearly visible in the right panel of Figure 6. For short horizons, the smaller-separation cases can have lower relative regret because the penalty for incorrect decisions is limited. As the horizon increases, however, the larger-separation cases become easier to learn and their regret decays more rapidly. In particular, the curve for $\epsilon_1 = \epsilon_2 = 0.25$ starts with the highest regret but becomes the best-performing case for large horizons. Thus, the experiment highlights the interaction between distinguishability and error cost in finite-horizon return-aware learning.

7.2. Experiment 2: Sensitivity Analysis

We next examine how the performance of COBRA changes with two economically important inputs: the return cost and the product price. In each experiment, we compare COBRA with the ETC benchmarks and MNL-Bandit on synthetically generated data.

Impact of return cost. The left panel of Figure 7 reports relative regret under low, intermediate, and high return-cost regimes. Relative regret increases for all methods as the return cost becomes larger, which is expected: when returns are more expensive, errors in identifying high-return products are penalized more heavily. COBRA remains clearly superior across all three regimes, and its performance is notably more stable than the benchmark policies. In particular, the gap between COBRA and MNL-Bandit widens as return costs rise, showing that return-aware learning becomes increasingly valuable when the operational consequences of returns are more severe.

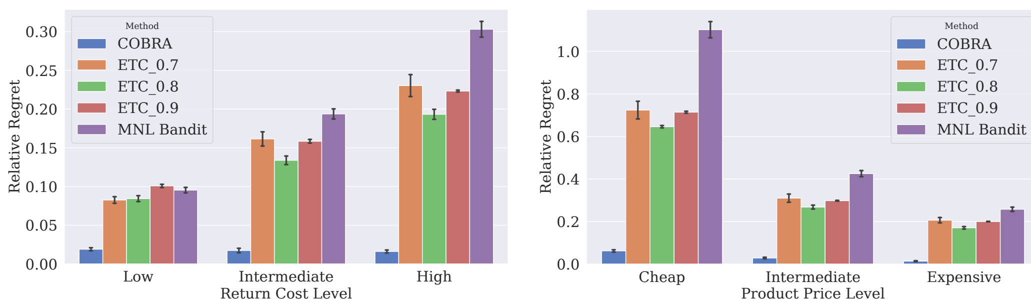

Figure 7 **Relative regret as a function of the return-cost level (left-panel) and the product-price level (right-panel). Error bars represent standard errors.**

Impact of product price. The right panel of Figure 7 reports relative regret under cheap, intermediate, and expensive price regimes. As prices increase, relative regret decreases for all methods. This pattern is intuitive in our setting: holding return costs fixed, higher prices make the return penalty smaller relative to the revenue collected from a successful sale, which reduces the severity of mistakes involving return-prone products. COBRA again performs best throughout, and its advantage over the benchmark policies is especially pronounced when prices are low, i.e., when returns are relatively most costly compared with revenue.

Taken together, these experiments show that COBRA is robust across a range of operational environments. The gains from explicitly modeling returns are largest when return costs are high or product prices are low, precisely the regimes in which ignoring returns is most damaging.

7.3. Experiment 3: Real-Data Study

We finally evaluate COBRA on real transaction and return data obtained from our retail collaborator, a fast-fashion retailer with a large online channel and frequent seasonal product introductions. As discussed in Section 1, this environment is particularly relevant for our problem: each season brings a large number of new products, many with little or no sales history at launch, while returns play a significant operational role due to the retailer's hassle-free return policy.

Our dataset contains SKU-level sales and return records from the retailer's online channel for a Fall–Winter season, together with product attributes such as prices and purchase costs. In the experiments below, we focus on three representative product categories: dresses, cardigans, and leggings. The partner retailer's return window is 30 days. We normalize prices to lie in $[0,1]$, compute the return cost of each item as the sum of its price and shipping cost, and set the assortment size to 10, matching the number of products displayed on the category landing page.

Using within-season transaction and return records, we estimate the product-level primitives and the return-delay distribution for each category, and then simulate the online decision problem under the estimated category-specific environment. The detailed estimation procedure is reported in Appendix EC.5. As benchmarks, we use ETC with three exploration durations, namely $T^{0.7}$, $T^{0.8}$, and $T^{0.9}$, together with MNL-Bandit.

Figure 8 reports the relative regret of all policies for the three categories, and Table 3 reports the corresponding total return volumes. The main conclusion is clear: COBRA achieves the lowest relative regret in all three categories. The advantage is especially pronounced relative to the ETC benchmarks, whose performance is much more sensitive to the choice of exploration length. MNL-Bandit is consistently the second-best policy in regret, but remains clearly worse than COBRA because it ignores return information in both estimation and assortment optimization.

The return-volume results provide a complementary operational perspective. COBRA reduces return volume relative to MNL-Bandit in all three categories, and it achieves the lowest total return volume in the cardigan category. In the dress and leggings categories, some ETC variants generate fewer total returns than COBRA, but they do so at the cost of substantially worse regret. This is consistent with the objective of our algorithm: COBRA is designed to optimize return-aware profit rather than return volume alone. The real-data results therefore reinforce the main message of the paper: explicitly modeling returns improves profit performance, and it often produces a meaningful side benefit in terms of lower return volume as well.

We also observe that the best ETC exploration length varies across categories. In the dress and cardigan categories, longer exploration performs better among the ETC variants, whereas in the leggings category the intermediate or shorter exploration choices are more competitive. This sensitivity highlights a practical weakness of ETC-type policies: their performance depends strongly on exploration tuning. By contrast, COBRA remains consistently strong across all three categories without category-specific retuning.

8. Conclusion

Motivated by a collaboration with a European fast-fashion retailer, this paper studies dynamic assortment learning with delayed and window-censored product returns. We develop a unified purchase–return model

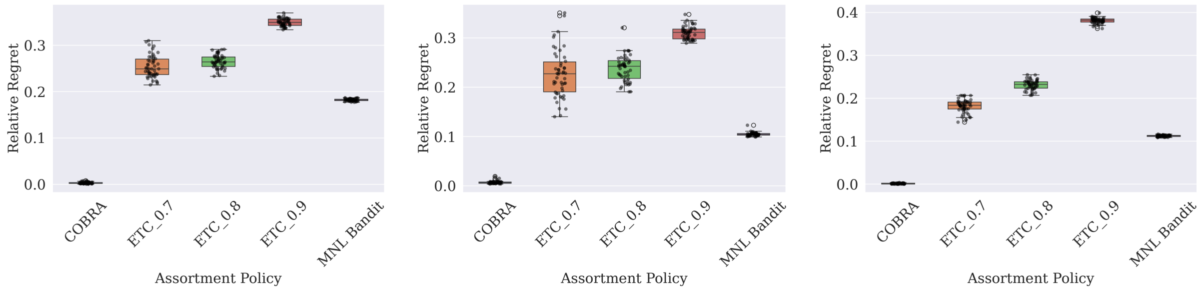

Figure 8 **Relative regret on the real-data experiments for the dress (left panel), cardigan (middle-panel), and leggings (right-panel) categories. Lower values are better.**

Table 3 **Total return volume on real-world datasets.**

Dataset:	Dress	Cardigan	Legging
MNL Bandit	4316	1602	2386
ETC $T^{0.7}$	3757	1615	2226
ETC $T^{0.8}$	3750	1666	2090
ETC $T^{0.9}$	3481	1594	1736
COBRA	3869	1410	2329

in which both the purchase decision and the subsequent keep/return decision are generated from common latent primitives. This allows the retailer to learn from both sales and returns while explicitly accounting for the fact that return information arrives only gradually over time.

We propose COBRA, an epoch-based return-aware learning algorithm that combines three components: product-level estimation of shared purchase–return primitives, estimation of the in-window return-delay profile, and optimistic assortment selection under joint uncertainty. We establish a regret upper bound that makes explicit two distinct sources of difficulty: the limited fraction of returns observable within the return window, and the censoring created by unresolved recent purchases. We also provide a lower bound showing that the leading $\sqrt{NT}$ dependence is unavoidable and that delayed return observability fundamentally affects the learning problem.

Our numerical results on both synthetic and real data show that explicitly incorporating returns can substantially improve profit performance relative to return-blind approaches. In addition, although return volume is not the optimization objective, the return-aware assortments produced by COBRA often lead to lower realized return volumes as a secondary operational benefit. The experiments also reveal a non-monotone role of the return window: expanding the window increases return observability, but it also prolongs the period during which recent purchases remain unresolved.

Several extensions remain open. An important next step is to incorporate richer contextual structure, such as product and customer features, into the joint purchase–return model. More broadly, our results suggest that returns should be treated not only as a cost, but also as a delayed information source that can improve dynamic merchandising decisions when modeled appropriately.

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EC.1. Technical Analysis of COBRA

This appendix collects the technical results underlying the analysis of COBRA. We proceed in the same order as the proof architecture in the main text. We first establish the purchase-side decoupling induced by the epoch structure, then study regularity and concentration properties of the per-product likelihood, quantify the bias introduced by delayed and censored returns, derive confidence bounds for both the product-level parameters and the delay profile, and finally prove the regret theorem.

EC.1.1. Purchase-side decoupling under the epoch structure

This subsection establishes the purchase-side decoupling induced by the epoch structure. Repeating a fixed assortment until the first no-purchase converts the MNL purchase process into per-product geometric observations, which is the key device underlying the product-wise likelihood construction used by COBRA.

This decoupling argument is standard in the MNL bandit literature; see, e.g., Agrawal et al. (2019). For completeness, we restate the result.

LEMMA EC.1. *Fix an epoch in which a nonempty assortment S is repeated until the first no-purchase. For any $i \in S$, let N_i be the number of purchases of i before stopping. Then N_i is geometric on* $\{0, 1, 2, \dots\}$ *with parameter*

$$\alpha_i(\theta_i) = \frac{1}{1 + \exp(q_i - \rho_i)}.$$

In particular, conditional on the sequence of assortments across epochs, $\{N_i\}$ is independent across epochs and its law depends on the model only through $q_i - \rho_i$.

This result allows the purchase component of the likelihood to be written in per-product form, which underlies the estimation step used by COBRA.

EC.1.2. Per-product likelihood regularity and curvature

We next establish smoothness and curvature properties of the penalized per-product likelihood. These results provide the deterministic regularity needed for concentration and parameter-error analysis.

LEMMA EC.2. *Fix a product i and an epoch index τ. Let*

$$\Theta_i := \{(q_i, \rho_i) : |q_i \pm \rho_i| \le \bar{u}\}, \qquad v := (1, -1)^\top, \qquad w := (1, 1)^\top,$$

so that $v^\top \theta_i = q_i - \rho_i$ and $w^\top \theta_i = q_i + \rho_i$. Let $\Psi(0) := 0$ and $\eta > 0$.

Let $(\mathcal{F}_t)_{t \ge 0}$ be the filtration generated by the algorithm and the observed purchase/return process up to time t. Assume S_t is $\mathcal{F}_{t-1}$-measurable. Let t_τ be the epoch boundary and define exposure counts

$$n_{i,\tau} := \sum_{t=1}^{t_\tau} \mathbf{1}\{i \in S_t\}, \qquad n_{i,\tau}^{\text{full}} := \sum_{t=1}^{t_\tau - \mu} \mathbf{1}\{i \in S_t\},$$

with $n_{i,\tau}^{\text{full}} = 0$ if $t_\tau \le \mu$.

Define

$$\alpha_i(\theta_i) := \frac{1}{1+\exp(v^\top \theta_i)}, \qquad r_i(\theta_i) := \frac{1}{1+\exp(w^\top \theta_i)}.$$

(i) *For fixed Ψ, the maps $\theta_i \mapsto \ell_{i,\tau}(\theta_i)$ and $\theta_i \mapsto \widetilde{\ell}_{i,\tau}(\theta_i)$ are twice continuously differentiable on Θ_i.*

(ii) *Let $H^{\mathrm{obs}}_{i,\tau}(\theta_i)$ and $H^{\mathrm{orc}}_{i,\tau}(\theta_i)$ denote the* expected negative Hessians *of the (penalized) per-product log-likelihoods:*

$$H^{\mathrm{obs}}_{i,\tau}(\theta_i) := -\mathbb{E}\big[\nabla^2_{\theta_i}\ell_{i,\tau}(\theta_i)\big] + \eta I_2, \qquad H^{\mathrm{orc}}_{i,\tau}(\theta_i) := -\mathbb{E}\Big[\nabla^2_{\theta_i}\widetilde{\ell}_{i,\tau}(\theta_i)\Big] + \eta I_2,$$

where I_2 is the 2×2 identity matrix and the expectation is taken over the customer randomness conditional on the (possibly random) assortment process $\{S_t\}_{t\le t_\tau}$ (equivalently, conditional on $\mathcal{F}_0$ and the policy's internal randomness). Then there exist constants $c_-, C_+ > 0$ depending only on $(\bar{u}, K, \eta)$ such that for all $\theta_i \in \Theta_i$,

$$c_-\,\psi_\mu\, n^{\mathrm{full}}_{i,\tau}\, I_2 \preceq H^{\mathrm{obs}}_{i,\tau}(\theta_i) \preceq C_+\, n_{i,\tau}\, I_2, \qquad c_-\,\psi_\mu\, n_{i,\tau}\, I_2 \preceq H^{\mathrm{orc}}_{i,\tau}(\theta_i) \preceq C_+\, n_{i,\tau}\, I_2.$$

In particular, since $n^{\mathrm{full}}_{i,\tau} \ge n_{i,\tau} - \mu$, if $n_{i,\tau} \ge 2\mu$ then

$$H^{\mathrm{obs}}_{i,\tau}(\theta_i) \succeq \frac{c_-}{2}\,\psi_\mu\, n_{i,\tau}\, I_2.$$

Proof. *Part (i).* On Θ_i, we have $|v^\top\theta_i| \le \bar{u}$ and $|w^\top\theta_i| \le \bar{u}$, hence $\alpha_i(\theta_i), r_i(\theta_i) \in (0,1)$ and are C^∞ in θ_i. The only θ_i-dependent terms appearing in $\ell_{i,\tau}$ and $\widetilde{\ell}_{i,\tau}$ are finite sums of $\log\alpha_i(\theta_i)$, $\log(1-\alpha_i(\theta_i))$, $\log r_i(\theta_i)$, and $\log(1 - c\, r_i(\theta_i))$ with $c \in [0,1]$ (specifically $c = \Psi(m)$ for some $m \le \mu$). Since $r_i(\theta_i) \le r_+ := e^{\bar{u}}/(1+e^{\bar{u}}) < 1$, we have $1 - c\, r_i(\theta_i) \ge 1 - r_+ > 0$ uniformly. Therefore both likelihoods are C^2 on Θ_i.

Part (ii). Fix $\theta_i \in \Theta_i$. We bound curvature separately in the orthogonal directions v and w.

Purchase direction (v). When $i \in S_t$, the per-epoch decoupling (Lemma EC.1) implies that the m-th purchase/no-purchase "relevant" trial for product i has log-likelihood contributions $\log\alpha_i(\theta_i)$ (no-purchase) and $\log(1-\alpha_i(\theta_i))$ (purchase of i), and trials involving other products contribute no θ_i-dependence. A direct calculation yields

$$-\frac{\partial^2}{\partial u^2}\log\alpha(u) = \alpha(u)(1-\alpha(u)), \qquad -\frac{\partial^2}{\partial u^2}\log(1-\alpha(u)) = \alpha(u)(1-\alpha(u)),$$

where $u = v^\top\theta_i$. Thus the expected negative Hessian contributed per exposure in direction v is proportional to $\alpha_i(\theta_i)(1-\alpha_i(\theta_i))\, vv^\top$, hence bounded between constants $a_- vv^\top$ and $a_+ vv^\top$ uniformly over Θ_i where $a_- := \sigma(-\overline{u})(1-\sigma(-\overline{u}))$ and $a_+ = 1/4$.

Return direction (w). Let $z = w^\top\theta_i$ and $r(z) = 1/(1+e^z)$. Conditional on a purchase of i, the "return within window" indicator Y satisfies $\Pr(Y=1) = c\, r(z)$ with $c \in [0, \psi_\mu]$, and the θ_i-dependent part of the marginal window log-likelihood is $Y \log r(z) + (1-Y)\log(1-c\,r(z))$ (constants in f_d dropped). A direct calculation shows the scalar Fisher information satisfies

$$\mathcal{I}_c(z) = -\mathbb{E}\left[\frac{\partial^2}{\partial z^2}\Big(Y \log r(z) + (1-Y)\log(1-c\,r(z))\Big)\right] = \frac{c\,r(z)\,(1-r(z))^2}{1-c\,r(z)}.$$

On Θ_i, $r(z) \in [r_-, r_+]$ with $r_- := 1/(1+e^{\bar{u}})$ and $r_+ := e^{\bar{u}}/(1+e^{\bar{u}})$, and $1 - c\,r(z) \in [1-r_+, 1] = [r_-, 1]$. Hence there exist $\iota_-, \iota_+ > 0$ depending only on $\bar{u}$ such that $\iota_- c \le \mathcal{I}_c(z) \le \iota_+ c$ for all $z \in [-\bar{u}, \bar{u}]$ and $c \in [0,1]$. Lifting back to θ_i, this contributes $\mathcal{I}_c(w^\top \theta_i)\, ww^\top$ per purchase.

Now, when $i \in S_t$, the purchase probability satisfies

$$\phi_{i,t} = \Pr(a_t = i \mid S_t) = \frac{e^{q_i - \rho_i}}{1 + \sum_{j \in S_t} e^{q_j - \rho_j}} \in [\phi_-, \phi_+],$$

with $\phi_- > 0$ depending only on $(\bar{u}, K)$ (using the global bounds $|q_j - \rho_j| \le \bar{u}$ and $|S_t| \le K$). Therefore the expected return-direction curvature per exposure is bounded between constants times $\psi_\mu\, ww^\top$.

In the oracle case, every exposure may generate a full-window observation with $c = \psi_\mu$, yielding a lower bound of order $\psi_\mu n_{i,\tau}$ in direction w. In the observed case at epoch boundary t_τ, restricting to times $t \le t_\tau - \mu$ ensures full windows, so the same bound holds with $n_{i,\tau}^{\text{full}}$.

Combine directions and add ridge. The purchase part yields $A\, vv^\top$ with $A \asymp n_{i,\tau}$, the return part yields $B\, ww^\top$ with $B \asymp \psi_\mu n_{i,\tau}$ (oracle) or $B \asymp \psi_\mu n_{i,\tau}^{\text{full}}$ (observed). Since $v^\top w = 0$ and $\|v\|_2^2 = \|w\|_2^2 = 2$, the eigenvalues along v and w directions scale as $2A$ and $2B$, and adding ηI_2 preserves positive definiteness. Absorbing fixed factors into constants yields the stated bounds.

□

The bounds in Lemma EC.2 show that the penalized likelihood has curvature of order exposure count, with the return component scaled by the observable in-window mass ψ_μ.

EC.1.3. Self-normalized concentration of the oracle score

This subsection controls the stochastic fluctuation of the oracle score process under adaptive data collection. The argument uses a self-normalized martingale bound, following the method-of-mixtures approach standard in the bandit literature.

LEMMA EC.3. *Fix a product i and an epoch τ. Let $(\widetilde{\mathcal{F}}_t)_{t \ge 0}$ be the enhanced natural filtration containing all purchase decisions up to time t and all return outcomes for items purchased at or before time t. Define per-period per-product contribution to oracle log-likelihood*

$$\widetilde{\ell}_{i,t}^{inc}(\theta_i) := \mathbb{I}\{i \in S_t\}\Big(\mathbb{I}\{a_t = 0\}\log \alpha_i(\theta_i) + \mathbb{I}\{a_t = i\}\Big[\log\big(1-\alpha_i(\theta_i)\big) + \sum_{\epsilon=1}^{\mu} X_{i,t,t+\epsilon}^{\mu}\big(\log r_i(\theta_i) + \log f_d(\epsilon)\big)$$
$$+ \big(1 - Z_{i,t}\big)\log\big(1 - r_i(\theta_i)\psi_\mu\big)\Big]\Big), \tag{EC.1}$$

such that

$$\widetilde{\ell}_{i,\tau}(\theta) = \sum_{t \le t_\tau} \widetilde{\ell}^{inc}_{i,t}(\theta).$$

Define the per-time score increment at the truth and the cumulative score

$$\xi_{i,t} := \nabla_{\theta_i} \widetilde{\ell}^{inc}_{i,t}(\theta^*) \in \mathbb{R}^2, \qquad g_{i,t} := \sum_{s \le t} \xi_{i,s}, \qquad g_i := g_{i,t_\tau}.$$

For each t, define the predictable Fisher increment at the truth

$$\Delta H^{(i)}_{*,t} := -\mathbb{E}\Big[\nabla^2_{\theta_i} \widetilde{\ell}^{inc}_{i,t}(\theta^*) \mid \widetilde{\mathcal{F}}_{t-1}\Big] \succeq 0, \qquad H^{(i)}_{*,t} := \sum_{s \le t} \Delta H^{(i)}_{*,s}.$$

Let $B_{i,t}$ be a predictable upper bound on the score norm $\|\xi_{i,t}\|_2$, such that $B_{i,t} := C_{\overline{u}} \cdot \mathbb{I}(i \in S_t)$ for a constant $C_{\overline{u}}$ depending on the compactness of Θ_i, and let

$$K^{(i)}_t := 2B^2_{i,t} I_2 + 2\Delta H^{(i)}_{*,t} \in \mathbb{S}^2_+, \qquad V^{(i)}_t := \sum_{s \le t} K^{(i)}_s = 2H^{(i)}_{*,t} + 2\Big(\sum_{s \le t} B^2_{i,s}\Big) I_2.$$

Fix any $V^{(i)}_0 \in \mathbb{S}^2_{++}$. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\|g_i\|^2_{(V^{(i)}_0 + V^{(i)}_{t_\tau})^{-1}} = g_i^\top (V^{(i)}_0 + V^{(i)}_{t_\tau})^{-1} g_i \;\le\; 2\ln\left(\frac{\det(V^{(i)}_0 + V^{(i)}_{t_\tau})^{1/2}}{\det(V^{(i)}_0)^{1/2}\,\delta}\right).$$

Equivalently,

$$\|g_i\|_{(V^{(i)}_0 + V^{(i)}_{t_\tau})^{-1}} \;\le\; \sqrt{2\ln\left(\frac{\det(V^{(i)}_0 + V^{(i)}_{t_\tau})^{1/2}}{\det(V^{(i)}_0)^{1/2}\,\delta}\right)}.$$

Proof. By the score property at the true parameter, $\mathbb{E}[\xi_{i,t} \mid \widetilde{\mathcal{F}}_{t-1}] = 0$ for all t, hence $(g_{i,t})_{t \ge 0}$ is a $\mathbb{R}^2$-valued martingale.

Recall that we defined a predictable bound $B_{i,t}$ on the score increment via

$$B_{i,t} := C_{\bar{u}}\, \mathbb{I}\{i \in S_t\},$$

where $C_{\bar{u}} < \infty$ is a uniform constant satisfying $\sup_{\theta_i \in \Theta_i} \|\nabla_{\theta_i} \widetilde{\ell}^{inc}_t(\theta)\|_2 \le C_{\bar{u}}$ (for all realizations of the binary outcomes in period t). In particular, $B_{i,t}$ is $\mathcal{F}_{t-1}$-measurable since $\mathbb{I}\{i \in S_t\}$ is, and pathwise $\|\xi_{i,t}\|_2 \le B_{i,t}$.

Fix $\lambda \in \mathbb{R}^2$ and define $X_{i,t}(\lambda) := \lambda^\top \xi_{i,t}$. Then $\mathbb{E}[X_{i,t}(\lambda) \mid \widetilde{\mathcal{F}}_{t-1}] = 0$ and

$$|X_{i,t}(\lambda)| \le \|\lambda\|_2\, \|\xi_{i,t}\|_2 \le B_{i,t}\, \|\lambda\|_2 \qquad \text{almost surely.}$$

By Hoeffding's lemma (conditional on $\widetilde{\mathcal{F}}_{t-1}$),

$$\log \mathbb{E}\Big[\exp(\lambda^\top \xi_{i,t}) \mid \widetilde{\mathcal{F}}_{t-1}\Big] \le \frac{B^2_{i,t}\|\lambda\|^2_2}{2}.$$

Since

$$\frac{1}{2}\lambda^\top K^{(i)}_t \lambda = \lambda^\top \big(B^2_{i,t} I_2 + \Delta H^{(i)}_{*,t}\big)\lambda \ge B^2_{i,t}\|\lambda\|^2_2,$$

we obtain for all $\lambda \in \mathbb{R}^2$,

$$\mathbb{E}\left[\exp\left(\lambda^\top \xi_{i,t} - \tfrac{1}{2}\lambda^\top K_t^{(i)} \lambda\right) \,\middle|\, \widetilde{\mathcal{F}}_{t-1}\right] \le 1. \tag{EC.2}$$

The conditional exponential moment bound (EC.2) is the condition used in the proof of Abbasi-Yadkori et al. (2011, Lemma 9). In particular (following the same Gaussian-mixture construction in their proof), define

$$\bar{V}_t^{(i)} := V_0^{(i)} + \sum_{s=1}^{t} K_s^{(i)}, \qquad g_{i,t} := \sum_{s=1}^{t} \xi_{i,s},$$

and consider the nonnegative process

$$\widetilde{M}_t^{(i)} := \left(\frac{\det(V_0^{(i)})}{\det(\bar{V}_t^{(i)})}\right)^{1/2} \exp\left(\frac{1}{2}\, g_{i,t}^\top (\bar{V}_t^{(i)})^{-1} g_{i,t}\right).$$

By the same completion-of-squares calculation as in the proof of Abbasi-Yadkori et al. (2011, Lemma 9), $(\widetilde{M}_t^{(i)})_{t\ge 0}$ is a supermartingale with $\mathbb{E}[\widetilde{M}_t^{(i)}] \le 1$ for all t. Therefore, by Ville's inequality,

$$\mathbb{P}\left(\sup_{t \le t_\tau} \widetilde{M}_t^{(i)} \ge \delta^{-1}\right) \le \delta.$$

Equivalently, with probability at least $1-\delta$, simultaneously for all $t \le t_\tau$,

$$\left(\frac{\det(V_0^{(i)})}{\det(\bar{V}_t^{(i)})}\right)^{1/2} \exp\left(\frac{1}{2}\, g_{i,t}^\top (\bar{V}_t^{(i)})^{-1} g_{i,t}\right) < \delta^{-1}.$$

Rearranging yields, for all $t \le t_\tau$,

$$g_{i,t}^\top (\bar{V}_t^{(i)})^{-1} g_{i,t} \;\le\; 2\ln\left(\frac{\det(\bar{V}_t^{(i)})^{1/2}}{\det(V_0^{(i)})^{1/2}\,\delta}\right),$$

which completes the proof.

□

Lemma EC.3 provides a time-uniform score bound for the oracle likelihood, which will later be combined with censoring-bias control to analyze the observed estimator.

EC.1.4. Bias induced by delayed and censored returns

The observed per-product likelihood used by COBRA is censored because, at epoch boundary t_τ, purchases made in the last μ periods may not yet have revealed their full return outcomes. To quantify the effect of this censoring, we compare the observed likelihood with an oracle benchmark in which every purchase is observed through the entire return window.

Fix a product i and epoch τ. Recall that

$$L_{i,\tau} := \left|\{\ell \le \tau : \; i \in S_\ell\}\right|$$

is the number of epochs up to t_τ in which product i is offered, and for each such epoch ℓ, $N_{i,\ell}$ is the number of purchases of i before the stopping no-purchase. For each purchase time $s \in \mathcal{B}_{i,\tau}$, define the observed window length

$$m_{s,\tau} := \min\{\mu,\ t_\tau - s\} \in \{0, 1, \dots, \mu\}.$$

At time t_τ, the observed return indicators associated with purchase s are

$$\{X^{\mu}_{i,s,s+\epsilon}\}_{\epsilon=1}^{m_{s,\tau}},$$

with at most one entry equal to 1.

The censored per-product likelihood is therefore

$$\mathcal{L}_{i,\tau}(\theta_i, \Psi) = \prod_{\ell=1}^{L_{i,\tau}} \Big[(1-\alpha_i(\theta_i))^{N_{i,\ell}}\, \alpha_i(\theta_i) \Big] \cdot \prod_{s \in \mathcal{B}_{i,\tau}} \left[\prod_{\epsilon=1}^{m_{s,\tau}} \big(r_i(\theta_i) f_d(\epsilon)\big)^{X^{\mu}_{i,s,s+\epsilon}} \right] \Big(1 - r_i(\theta_i)\Psi(m_{s,\tau})\Big)^{1-\sum_{\epsilon=1}^{m_{s,\tau}} X^{\mu}_{i,s,s+\epsilon}}, \tag{EC.3}$$

$$\ell_{i,\tau}(\theta_i, \Psi) := \log \mathcal{L}_{i,\tau}(\theta_i, \Psi).$$

For the analysis, we also introduce the oracle likelihood in which every purchase is observed through the full return window of length μ. Specifically, for each $s \in \mathcal{B}_{i,\tau}$ we set $\tilde{m}_{s,\tau} := \mu$ and observe

$$\{X^{\mu}_{i,s,s+\epsilon}\}_{\epsilon=1}^{\mu}.$$

The corresponding oracle per-product likelihood is

$$\widetilde{\mathcal{L}}_{i,\tau}(\theta_i, \Psi) = \prod_{\ell=1}^{L_{i,\tau}} \Big[(1-\alpha_i(\theta_i))^{N_{i,\ell}}\, \alpha_i(\theta_i) \Big] \cdot \prod_{s \in \mathcal{B}_{i,\tau}} \left[\prod_{\epsilon=1}^{\mu} \big(r_i(\theta_i) f_d(\epsilon)\big)^{X^{\mu}_{i,s,s+\epsilon}} \right] \Big(1 - r_i(\theta_i)\Psi(\mu)\Big)^{1-\sum_{\epsilon=1}^{\mu} X^{\mu}_{i,s,s+\epsilon}}, \tag{EC.4}$$

$$\widetilde{\ell}_{i,\tau}(\theta_i, \Psi) := \log \widetilde{\mathcal{L}}_{i,\tau}(\theta_i, \Psi).$$

COBRA can compute $\ell_{i,\tau}$ from data available by time t_τ, but cannot compute $\widetilde{\ell}_{i,\tau}$ because the latter depends on return outcomes in $(t_\tau, t_\tau + \mu]$ for purchases made in the last μ periods. Consequently, the discrepancy between the observed and oracle likelihoods is entirely driven by unresolved recent purchases. The next lemma quantifies the resulting score bias and shows that it can be controlled by a term depending on the return window length and a plug-in term arising from delay-profile estimation.

Lemma EC.4. *For any product i and epoch τ, let $b_i := \nabla_i \ell_\tau(\theta^*; \hat{\Psi}) - \nabla_i \tilde{\ell}_\tau(\theta^*; \Psi)$ be the score bias between the censored and ideal log-likelihoods. Under Assumption 2, there exist constants $\kappa_3, \kappa_4, \kappa_\psi > 0$ such that, with probability at least $1 - N^{-2}T^{-2}$,*

$$\|b_i\|_2 \le \min\Big\{\mu,\ \kappa_3\sqrt{\mu \ln(NT)} + \kappa_4 \overline{d}\Big\} + \kappa_\psi \psi_\mu^{-1}\, n_{i,\tau} \max_{m \le \mu} \big|\widehat{\Psi}^{\tau}(m) - \Psi(m)\big|.$$

Proof. Let $\mathcal{B}_{i,\tau}$ be the set of purchase times of product i up to t_τ, and define the subset of *recent* purchases whose return window has not yet closed:

$$\mathcal{B}_{i,\mathrm{rec}} := \{s \in \mathcal{B}_{i,\tau} : t_\tau - s < \mu\}.$$

For any purchase $s \in \mathcal{B}_{i,\tau}$ with $t_\tau - s \geq \mu$, the full μ-window has elapsed, so the censored and full log-likelihood contributions coincide and do not contribute to the bias arising from incomplete windows. Hence, if we temporarily *freeze* the return delay distribution at its true value ψ_μ and write

$$\ell_\tau(\theta) := \ell_\tau(\theta; \Psi), \qquad \tilde{\ell}_\tau(\theta) := \tilde{\ell}_\tau(\theta; \Psi).$$

Let

$$\ell^{inc}_{i,t,\tau}(\theta_i) := \mathbb{I}\{i \in S_t\}\Big(\mathbb{I}\{a_t = 0\} \log \alpha_i(\theta_i) + \mathbb{I}\{a_t = i\}\Big[\log\big(1 - \alpha_i(\theta_i)\big) \tag{EC.5}$$

$$+ \sum_{\epsilon=1}^{m_{t,\tau}} X^{\mu}_{i,t,t+\epsilon}\big(\log r_i(\theta_i) + \log f_d(\epsilon)\big) + \big(1 - Z^{(m_{t,\tau})}_{i,t}\big) \log\big(1 - r_i(\theta_i)\Psi(m_{t,\tau})\big)\Big]\Big). \tag{EC.6}$$

and

$$\widetilde{\ell}^{inc}_{i,t}(\theta_i) := \mathbb{I}\{i \in S_t\}\Big(\mathbb{I}\{a_t = 0\} \log \alpha_i(\theta_i) + \mathbb{I}\{a_t = i\}\Big[\log\big(1 - \alpha_i(\theta_i)\big) + \sum_{\epsilon=1}^{\mu} X^{\mu}_{i,t,t+\epsilon}\big(\log r_i(\theta_i) + \log f_d(\epsilon)\big)$$

$$+ \big(1 - Z_{i,t}\big) \log\big(1 - r_i(\theta_i)\psi_\mu\big)\Big]\Big), \tag{EC.7}$$

Then the corresponding "window-bias" is

$$b^{\mathrm{win}}_i := \nabla_i \ell_\tau(\theta^*) - \nabla_i \tilde{\ell}_\tau(\theta^*) = \sum_{s \in \mathcal{B}_{i,\mathrm{rec}}} \Big(\nabla_i \ell^{inc}_s(\theta^*) - \nabla_i \tilde{\ell}^{inc}_s(\theta^*)\Big) =: \sum_{s \in \mathcal{B}_{i,\mathrm{rec}}} \Delta_{i,s},$$

where $\Delta_{i,s}$ is the gradient difference of ℓ^{inc}_s and $\tilde{\ell}^{inc}_s$ for product i.

By the model, there is at most one purchase per period, so $|\mathcal{B}_{i,\mathrm{rec}}| \leq \mu$.

The per-purchase score difference $\Delta_{i,s}$ is a measurable function of $(q_i, \rho_i) \in \Theta_i$ and the binary indicators in the μ-window. By compactness of Θ_i, all choice/return probabilities appearing in the log-likelihood are uniformly bounded away from 0, hence the per-purchase gradients are uniformly bounded. Therefore all log-arguments in ℓ^{inc} and $\tilde{\ell}^{inc}$ are uniformly bounded away from 0, and the corresponding gradients in (q_i, ρ_i) are uniformly bounded. Consequently, there exists a deterministic constant $\kappa_0 > 0$ (depending only on Θ_i and K) such that, almost surely,

$$\|\Delta_{i,s}\|_2 = \big\|\nabla_i \ell^{inc}_s(\theta^*) - \nabla_i \tilde{\ell}^{inc}_s(\theta^*)\big\|_2 \leq \kappa_0 \qquad \text{for all } s.$$

By the triangle inequality:

$$\|b^{\mathrm{win}}_i\|_2 \leq \sum_{s \in \mathcal{B}_{i,\mathrm{rec}}} \|\Delta_{i,s}\|_2 \leq \kappa_0 |\mathcal{B}_{i,\mathrm{rec}}| \leq \kappa_0 \mu. \tag{EC.8}$$

We now seek a sharper high-probability bound when μ is large. Define

$$\Delta_{i,s} := \nabla_i \ell_s^{inc}(\theta^*) - \nabla_i \tilde{\ell}_s^{inc}(\theta^*), \quad b_i^{\text{win}} = \sum_{s \in \mathcal{B}_{i,\text{rec}}} \Delta_{i,s}.$$

Write

$$b_i^{\text{win}} = \left(b_i^{\text{win}} - \mathbb{E}[b_i^{\text{win}}]\right) + \mathbb{E}[b_i^{\text{win}}] \tag{EC.9}$$

$$= \sum_{s \in \mathcal{B}_{i,\text{rec}}} \left(\Delta_{i,s} - \mathbb{E}[\Delta_{i,s}]\right) + \sum_{s \in \mathcal{B}_{i,\text{rec}}} \mathbb{E}[\Delta_{i,s}]. \tag{EC.10}$$

We will bound the fluctuation term using a concentration inequality for independent bounded vectors, and the expectation term using the tail of the delay distribution.

For each $s \in \mathcal{B}_{i,\text{rec}}$, define

$$Y_s := \Delta_{i,s} - \mathbb{E}[\Delta_{i,s}].$$

Under the model, return delays of distinct purchases are independent, and each $\Delta_{i,s}$ depends only on the outcomes associated with purchase s (and the censoring induced by t_τ). Hence, conditional on $\mathcal{B}_{i,\text{rec}}$, the collection $\{Y_s\}_{s \in \mathcal{B}_{i,\text{rec}}}$ consists of independent, mean-zero random vectors.

Moreover, there exists a universal constant $C_1 > 0$ such that

$$\|Y_s\|_2 \le \|\Delta_{i,s}\|_2 + \|\mathbb{E}[\Delta_{i,s}]\|_2 \le C_1 \quad \text{for all } s.$$

Let $S := \sum_{s \in \mathcal{B}_{i,\text{rec}}} Y_s$. For each coordinate $k = 1, 2$, we have

$$\mathbb{P}\big(|\langle e_k, S\rangle| \ge x \,\big|\, \mathcal{B}_{i,\text{rec}}\big) \le 2\exp\left(-\frac{x^2}{2C_1^2 |\mathcal{B}_{i,\text{rec}}|}\right)$$

by Hoeffding's inequality. Using $\|S\|_2^2 = S_1^2 + S_2^2$ and the bound

$$\{\|S\|_2 \ge u\} \subseteq \left\{\max_{k=1,2} |S_k| \ge \tfrac{u}{\sqrt{2}}\right\},$$

a union bound over $k = 1, 2$, and $|\mathcal{B}_{i,\text{rec}}| \le \mu$ gives

$$\mathbb{P}\left(\left\|b_i^{\text{win}} - \mathbb{E}[b_i^{\text{win}}]\right\|_2 \ge u\right) \le 4\exp\left(-\frac{u^2}{4C_1^2 \mu}\right), \quad u > 0.$$

Choose

$$u = 2C_1\sqrt{\mu \ln\left(4N^2T^2\right)}$$

to obtain

$$\mathbb{P}\left(\left\|b_i^{\text{win}} - \mathbb{E}[b_i^{\text{win}}]\right\|_2 \ge 2C_1\sqrt{\mu \ln(4N^2T^2)}\right) \le (NT)^{-2}. \tag{EC.11}$$

Using $\ln(4N^2T^2) \le 3\ln(NT)$ for $N, T \ge 2$ and absorbing constants into a universal $\kappa_3 > 0$, we conclude that, with probability at least $1 - N^{-2}T^{-2}$,

$$\left\|b_i^{\text{win}} - \mathbb{E}[b_i^{\text{win}}]\right\|_2 \le \kappa_3\sqrt{\mu\ln(NT)}.$$

We now bound $\|\mathbb{E}[b_i^{\text{win}}]\|_2$. Fix $s \in \mathcal{B}_{i,\text{rec}}$ and let d_s be the return delay for the unit purchased at time s (if it ever returns). The gradient difference $\Delta_{i,s}$ arises from the fact that, at time t_τ, we do not yet know whether a late return will occur after t_τ. In particular, the contribution is non-zero only when $d_s > t_\tau - s$ (i.e., a "tail" return beyond the current cutoff), which has probability $\mathbb{P}(d_s > t_\tau - s)$.

The per-purchase gradient contribution is uniformly bounded, so there exists a universal constant $\kappa_4 > 0$ such that

$$\left\|\mathbb{E}[\Delta_{i,s}]\right\|_2 \le \mathbb{E}\big[\|\Delta_{i,s}\|_2\big] \le \kappa_4\,\mathbb{P}(d_s > t_\tau - s).$$

Hence

$$\begin{aligned}\left\|\mathbb{E}[b_i^{\text{win}}]\right\|_2 &= \Big\|\sum_{s\in\mathcal{B}_{i,\text{rec}}}\mathbb{E}[\Delta_{i,s}]\Big\|_2 \le \sum_{s\in\mathcal{B}_{i,\text{rec}}}\left\|\mathbb{E}[\Delta_{i,s}]\right\|_2 \\ &\le \kappa_4\sum_{s\in\mathcal{B}_{i,\text{rec}}}\mathbb{P}(d_s > t_\tau - s).\end{aligned}$$

Let $m := |\mathcal{B}_{i,\text{rec}}| \le \mu$. Order the purchases in $\mathcal{B}_{i,\text{rec}}$ by recency and write their ages as

$$1 \le a_1 < a_2 < \cdots < a_m < \mu, \qquad a_j := t_\tau - s_j.$$

Since the a_j are distinct positive integers, $a_j \ge j$ for all j. Because $k \mapsto \Pr(d > k)$ is nonincreasing, it follows that

$$\Pr(d_{s_j} > t_\tau - s_j) \;=\; \Pr(d > a_j) \;\le\; \Pr(d > j), \qquad j = 1, \ldots, m.$$

Therefore

$$\left\|\mathbb{E}[b_i^{\text{win}}]\right\|_2 \le \kappa_4\sum_{k=1}^{|\mathcal{B}_{i,\text{rec}}|}\mathbb{P}(d > k) \le \kappa_4\sum_{k=1}^{\mu}\mathbb{P}(d > k) \le \kappa_4\sum_{k=1}^{\infty}\mathbb{P}(d > k).$$

For any integer-valued nonnegative random variable d, we have

$$\sum_{k=1}^{\infty}\mathbb{P}(d > k) \le \mathbb{E}[d].$$

By Assumption 2, $\mathbb{E}[d] \le \overline{d}$, hence

$$\left\|\mathbb{E}[b_i^{\text{win}}]\right\|_2 \le \kappa_4\overline{d}.$$

On the event of (EC.11) (which has probability at least $1 - N^{-2}T^{-2}$), we have

$$\begin{aligned}\|b_i^{\text{win}}\|_2 &\le \left\|b_i^{\text{win}} - \mathbb{E}[b_i^{\text{win}}]\right\|_2 + \left\|\mathbb{E}[b_i^{\text{win}}]\right\|_2 \\ &\le \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\overline{d}.\end{aligned}$$

Combining with the deterministic bound from (EC.8), we obtain

$$\|b_i^{\text{win}}\|_2 \le \min\left\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\bar{d}\right\}.$$

In the actual algorithm, the censored log-likelihood used to define the score block for θ_i is evaluated at the plug-in estimate $\widehat{\Psi}^\tau(\cdot)$. The total bias in the profile score can then be written as

$$\begin{aligned} b_i &= \nabla_i\ell_\tau(\theta^*;\hat{\Psi}) - \nabla_i\tilde{\ell}_\tau(\theta^*;\Psi) \\ &= \underbrace{\Big(\nabla_i\ell_\tau(\theta^*;\Psi) - \nabla_i\tilde{\ell}_\tau(\theta^*;\Psi)\Big)}_{b_i^{\text{win}}} + \underbrace{\Big(\nabla_i\ell_\tau(\theta^*;\hat{\Psi}) - \nabla_i\ell_\tau(\theta^*;\Psi)\Big)}_{b_i^{(\Psi)}}. \end{aligned}$$

The first term b_i^{win} is exactly the quantity bounded above.

For the plug-in term $b_i^{(\Psi)}$, apply a mean-value expansion along the segment between Ψ and $\widehat{\Psi}^\tau$: there exists $\tilde{\Psi}$ on this segment such that

$$b_i^{(\Psi)} = \sum_{m=1}^{\mu} \partial^2_{\theta_i\Psi(m)}\ell_\tau(\theta^*,\tilde{\Psi})\left(\widehat{\Psi}^\tau(m) - \Psi(m)\right).$$

Hence, by the triangle inequality,

$$\|b_i^{(\Psi)}\|_2 \le \left(\max_{m\le\mu}\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right|\right)\sum_{m=1}^{\mu}\left\|\partial^2_{\theta_i\Psi(m)}\ell_\tau(\theta^*,\tilde{\Psi})\right\|_2.$$

Let $m_{i,\tau}(m)$ denote the number of epoch-τ likelihood terms involving product i whose contribution depends on the coordinate $\Psi(m)$; note that $\sum_{m=1}^{\mu} m_{i,\tau}(m) \le n_{i,\tau}$. Moreover, the cross-derivative is uniformly bounded as

$$\left\|\partial^2_{\theta_i\Psi(m)}\ell_\tau(\theta^*,\tilde{\Psi})\right\|_2 \le \kappa_\psi\,\psi_\mu^{-1}\,m_{i,\tau}(m), \qquad m = 1,\dots,\mu.$$

Therefore,

$$\sum_{m=1}^{\mu}\left\|\partial^2_{\theta_i\Psi(m)}\ell_\tau(\theta^*,\tilde{\Psi})\right\|_2 \le \kappa_\psi\,\psi_\mu^{-1}\sum_{m=1}^{\mu} m_{i,\tau}(m) \le \kappa_\psi\,\psi_\mu^{-1}\,n_{i,\tau},$$

and thus

$$\|b_i^{(\Psi)}\|_2 \le \kappa_\psi\,\psi_\mu^{-1}\,n_{i,\tau}\max_{m\le\mu}\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right|.$$

Finally, by the triangle inequality,

$$\|b_i\|_2 \le \|b_i^{\text{win}}\|_2 + \|b_i^{(\Psi)}\|_2 \le \min\left\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\bar{d}\right\} + \kappa_\psi\,\psi_\mu^{-1}\,n_{i,\tau}\max_{m\le\mu}\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right|.$$

This holds on the same event of probability at least $1 - N^{-2}T^{-2}$.

□

The resulting bias decomposition separates a censoring term due to unresolved recent purchases from a plug-in term due to delay-profile estimation. This is the key step through which the return window length μ enters the regret bound.

EC.1.5. Estimation of the delay profile

We now study the estimator of the delay CDF profile used by COBRA. Since the estimator employs a plug-in normalizer based on the current return-probability estimates, its analysis requires controlling both empirical fluctuation and denominator mismatch.

LEMMA EC.5. *There exist universal constants $c_1, c_2 > 0$ such that, for every epoch τ, with probability at least $1 - O(N^{-2}T^{-2})$, the following holds* simultaneously for all $m \in \{1, \dots, \mu\}$*:*

$$\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right| \le \beta_\tau := c_1\sqrt{\frac{\ln(NT\mu)}{S_\tau^{\text{full}}}} + c_2\frac{\ln(NT\mu)}{S_\tau^{\text{full}}} + \frac{\sum_{(i,s)\in P_\tau^{\text{full}}}\delta_{i,\tau-1}}{S_\tau^{\text{full}}},$$

whenever the plug-in errors satisfy

$$\left|r_i(\theta_i) - r_i(\hat{\theta}_{i,\tau-1})\right| \le \delta_{i,\tau-1} \qquad \text{for all } (i,s) \in P_\tau^{\text{full}}.$$

In particular, letting $\psi_\mu := \Psi(\mu)$ and $\widehat{\psi}_\mu^\tau := \widehat{\Psi}^\tau(\mu)$ gives the same bound for $\left|\widehat{\psi}_\mu^\tau - \psi_\mu\right|$.

Proof. For notational brevity, all sums over (i,s) in this proof are over $(i,s) \in P_\tau^{\text{full}}$. Fix any $m \in \{1, \dots, \mu\}$.

Define the (unclipped) ratio estimator

$$\widetilde{\Psi}^\tau(m) := \frac{1}{S_\tau^{\text{full}}}\sum_{(i,s)} Z_{i,s}^{\le m}, \qquad \widehat{\Psi}^\tau(m) := \left[\widetilde{\Psi}^\tau(m)\right]_{[0,1]}.$$

Since $\Psi(m) \in [0,1]$ and clipping is non-expansive on $[0,1]$, we have

$$\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right| \le \left|\widetilde{\Psi}^\tau(m) - \Psi(m)\right|.$$

Thus it suffices to bound $\left|\widetilde{\Psi}^\tau(m) - \Psi(m)\right|$.

By the model, for each matured purchase (i,s),

$$Z_{i,s}^{\le m} \sim \text{Bernoulli}(p_i(m)) \quad \text{with} \quad p_i(m) = r_i(\theta_i)\,\Psi(m),$$

and the $\{Z_{i,s}^{\le m}\}$ are independent across (i,s). Hence

$$\mathbb{E}\Big[\sum_{(i,s)} Z_{i,s}^{\le m}\Big] = \Psi(m)\sum_{(i,s)} r_i(\theta_i).$$

We can write

$$\begin{aligned}
\widetilde{\Psi}^\tau(m) - \Psi(m) &= \frac{1}{S_\tau^{\text{full}}}\sum_{(i,s)} Z_{i,s}^{\le m} - \Psi(m) \\
&= \frac{1}{S_\tau^{\text{full}}}\sum_{(i,s)}\left(Z_{i,s}^{\le m} - \mathbb{E}[Z_{i,s}^{\le m}]\right) + \frac{1}{S_\tau^{\text{full}}}\sum_{(i,s)}\mathbb{E}[Z_{i,s}^{\le m}] - \Psi(m) \\
&= \frac{1}{S_\tau^{\text{full}}}\sum_{(i,s)}\left(Z_{i,s}^{\le m} - \mathbb{E}[Z_{i,s}^{\le m}]\right) + \Psi(m)\Big(\frac{\sum_{(i,s)} r_i(\theta_i)}{S_\tau^{\text{full}}} - 1\Big).
\end{aligned}$$

Taking absolute values and using $\Psi(m) \le 1$ yields

$$\left|\widetilde{\Psi}^{\tau}(m) - \Psi(m)\right| \le \frac{\left|\sum_{(i,s)} (Z_{i,s}^{\le m} - \mathbb{E}[Z_{i,s}^{\le m}])\right|}{S_\tau^{\text{full}}} + \frac{\left|\sum_{(i,s)} r_i(\theta_i) - S_\tau^{\text{full}}\right|}{S_\tau^{\text{full}}}. \tag{EC.12}$$

Second term (denominator mismatch). Using $S_\tau^{\text{full}} = \sum_{(i,s)} r_i(\hat{\theta}_{i,\tau-1})$,

$$\begin{aligned}\Big|\sum_{(i,s)} r_i(\theta_i) - S_\tau^{\text{full}}\Big| &= \Big|\sum_{(i,s)} \big(r_i(\theta_i) - r_i(\hat{\theta}_{i,\tau-1})\big)\Big| \\ &\le \sum_{(i,s)} \big|r_i(\theta_i) - r_i(\hat{\theta}_{i,\tau-1})\big| \le \sum_{(i,s)} \delta_{i,\tau-1}.\end{aligned}$$

Therefore,

$$\frac{\left|\sum_{(i,s)} r_i(\theta_i) - S_\tau^{\text{full}}\right|}{S_\tau^{\text{full}}} \le \frac{\sum_{(i,s)} \delta_{i,\tau-1}}{S_\tau^{\text{full}}}.$$

First term (fluctuation of the centered sum). Consider centered bounded variables

$$X_{i,s}^{(m)} := Z_{i,s}^{\le m} - \mathbb{E}[Z_{i,s}^{\le m}], \qquad |X_{i,s}^{(m)}| \le 1.$$

Let

$$V(m) := \sum_{(i,s)} \mathrm{Var}(Z_{i,s}^{\le m}) = \sum_{(i,s)} \mathrm{Var}(X_{i,s}^{(m)}).$$

Since $Z_{i,s}^{\le m}$ is Bernoulli,

$$\mathrm{Var}(Z_{i,s}^{\le m}) \le \mathbb{E}[Z_{i,s}^{\le m}] = r_i(\theta_i)\Psi(m) \le r_i(\theta_i),$$

so

$$V(m) \le \sum_{(i,s)} r_i(\theta_i).$$

Apply Bernstein's inequality to $\sum X_{i,s}^{(m)}$. For any $\alpha \in (0,1)$, with probability at least $1-\alpha$,

$$\Big|\sum_{(i,s)} X_{i,s}^{(m)}\Big| \le \sqrt{2\ln(2/\alpha)\,V(m)} + \frac{\ln(2/\alpha)}{3}.$$

Moreover, by the plug-in condition and $|r_i(\theta_i) - r_i(\hat{\theta}_{i,\tau-1})| \le \delta_{i,\tau-1}$,

$$\sum_{(i,s)} r_i(\theta_i) \le S_\tau^{\text{full}} + \sum_{(i,s)} \delta_{i,\tau-1}.$$

Compactness implies $r_i(\hat{\theta}_{i,\tau-1}) \ge r_{\min} > 0$ uniformly, hence $|P_\tau^{\text{full}}| \le r_{\min}^{-1} S_\tau^{\text{full}}$ and with $\delta_{i,\tau-1} \le 1$ we get $\sum_{(i,s)} \delta_{i,\tau-1} \le C\, S_\tau^{\text{full}}$ for a universal C. Therefore, for some universal $c_V > 0$,

$$V(m) \le S_\tau^{\text{full}} + \sum_{(i,s)} \delta_{i,\tau-1} \le c_V^2\, S_\tau^{\text{full}}.$$

Consequently,

$$\frac{\left|\sum_{(i,s)} X_{i,s}^{(m)}\right|}{S_\tau^{\text{full}}} \le c_V \sqrt{\frac{2\ln(2/\alpha)}{S_\tau^{\text{full}}}} + \frac{\ln(2/\alpha)}{3 S_\tau^{\text{full}}}.$$

Plugging into (EC.12) yields: with probability at least $1-\alpha$,

$$\left|\widetilde{\Psi}^{\tau}(m)-\Psi(m)\right| \le c_V\sqrt{\frac{2\ln(2/\alpha)}{S_{\tau}^{\text{full}}}}+\frac{\ln(2/\alpha)}{3S_{\tau}^{\text{full}}}+\frac{\sum_{(i,s)}\delta_{i,\tau-1}}{S_{\tau}^{\text{full}}}.$$

Since $\left|\widehat{\Psi}^{\tau}(m)-\Psi(m)\right| \le \left|\widetilde{\Psi}^{\tau}(m)-\Psi(m)\right|$, the same bound holds for $\widehat{\Psi}^{\tau}(m)$.

Finally, to control the error simultaneously for all $m \in \{1,\dots,\mu\}$, we take a union bound. Set $\alpha := \frac{(NT)^{-2}}{\mu}$. Then the probability that there exists any m violating the bound is at most $\mu\alpha = (NT)^{-2}$. The log-factor becomes:

$$\ln(2/\alpha) = \ln(2\mu N^2T^2) \le C\ln(NT\mu),$$

for a universal constant C. Thus, there exist universal constants $c_1, c_2 > 0$ such that

$$c_V\sqrt{\frac{2\ln(2/\alpha)}{S_{\tau}^{\text{full}}}} \le c_1\sqrt{\frac{\ln(NT\mu)}{S_{\tau}^{\text{full}}}}, \qquad \frac{\ln(2/\alpha)}{3S_{\tau}^{\text{full}}} \le c_2\frac{\ln(NT\mu)}{S_{\tau}^{\text{full}}}.$$

This yields the claimed inequality simultaneously for all m with probability at least $1-(NT)^{-2}$. □

Lemma EC.5 yields a uniform bound on the estimated delay profile, and in particular on the scalar in-window mass ψ_μ used in the optimistic selection step.

EC.1.6. Uniform Hessian control for the censored objective

To convert score control into parameter-error control, we require a uniform approximation of the observed Hessian by its predictable counterpart. We first introduce the notation needed for the decomposition of the censored likelihood.

Fix a product i and an epoch boundary time t_τ. Recall that the penalized objective is

$$f_{i,\tau}(\theta_i) := \ell_{i,\tau}(\theta_i,\widehat{\Psi}) - \frac{\eta}{2}\|\theta_i\|_2^2, \qquad \eta > 0.$$

Let $n_{i,\tau} := |\mathcal{A}_{i,\tau}|$ denote the exposure count and write $n := n_{i,\tau} \vee 1$. For each purchase time $s \in \mathcal{B}_{i,\tau}$ define

$$m_{s,\tau} := \min\{\mu, t_\tau - s\}, \qquad Z_{i,s}^{(m)} := \sum_{\epsilon=1}^{m} X_{i,s,s+\epsilon}^{\mu}.$$

We partition purchase times into

$$\mathcal{B}_{i,\tau}^{\text{full}} := \{s \in \mathcal{B}_{i,\tau} : t_\tau - s \ge \mu\}, \qquad \mathcal{B}_{i,\tau}^{\text{rec}} := \{s \in \mathcal{B}_{i,\tau} : t_\tau - s < \mu\}.$$

Correspondingly, the θ_i-dependent component of the likelihood can be decomposed as

$$\ell_{i,\tau}(\theta_i,\widehat{\Psi}) = \ell_{i,\tau}^{\text{purch}}(\theta_i) + \ell_{i,\tau}^{\text{ret,full}}(\theta_i) + \ell_{i,\tau}^{\text{ret,rec}}(\theta_i) + (\text{terms independent of } \theta_i),$$

where

$$\ell_{i,\tau}^{\text{purch}}(\theta_i) = \sum_{\ell=1}^{L_{i,\tau}}\Big[N_{i,\ell}\log(1-\alpha_i(\theta_i)) + \log\alpha_i(\theta_i)\Big],$$

$$\ell_{i,\tau}^{\text{ret,full}}(\theta_i) = \sum_{s\in\mathcal{B}_{i,\tau}^{\text{full}}} \Big[Z_{i,s}^{(\mu)} \log r_i(\theta_i) + (1 - Z_{i,s}^{(\mu)}) \log\!\big(1 - r_i(\theta_i)\widehat{\Psi}^{\tau}(\mu)\big) \Big],$$

$$\ell_{i,\tau}^{\text{ret,rec}}(\theta_i) = \sum_{s\in\mathcal{B}_{i,\tau}^{\text{rec}}} \Big[Z_{i,s}^{(m_{s,\tau})} \log r_i(\theta_i) + (1 - Z_{i,s}^{(m_{s,\tau})}) \log\!\big(1 - r_i(\theta_i)\widehat{\Psi}^{\tau}(m_{s,\tau})\big) \Big].$$

Define the partially observed objective

$$f_{i,\tau}^{\text{full}}(\theta_i) := \ell_{i,\tau}^{\text{purch}}(\theta_i) + \ell_{i,\tau}^{\text{ret,full}}(\theta_i) - \frac{\eta}{2}\|\theta_i\|_2^2.$$

Let the enlarged filtration be

$$\mathcal{G}_t := \sigma(\mathcal{F}_t, \widehat{\Psi}^{\tau}),$$

so that $\widehat{\Psi}^{\tau}$ is $\mathcal{G}_{t-1}$–measurable.

We define the predictable negative Hessian

$$H_{i,\tau}^{\text{full}}(\theta_i) := -\sum_{t=1}^{t_\tau} \mathbb{E}\big[\nabla_{\theta_i}^2 \Delta f_{i,t}^{\text{full}}(\theta_i) \,\big|\, \mathcal{G}_{t-1}\big],$$

where $\Delta f_{i,t}^{\text{full}}$ denotes the per-time contribution of $f_{i,\tau}^{\text{full}}$.

LEMMA EC.6. *Let Θ_i be the compact parameter set and define*

$$r_- := \frac{1}{1+e^{\bar{u}}}, \qquad D := \sup_{\theta,\theta'\in\Theta_i} \|\theta - \theta'\|_2, \qquad n := n_{i,\tau} \vee 1.$$

Then there exist constants $C_1 = C_1(\bar{u},\eta) > 0$ and $C_2 = C_2(\bar{u}) > 0$ such that for any $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\sup_{\theta_i\in\Theta_i} \big\| -\nabla_{\theta_i}^2 f_{i,\tau}(\theta_i) - H_{i,\tau}^{\text{full}}(\theta_i) \big\|_{\text{op}} \le C_1\Big(\sqrt{n_{i,\tau}\,\Lambda_{i,\tau}(\delta)} + \Lambda_{i,\tau}(\delta)\Big) + C_2\,\mu,$$

where

$$\Lambda_{i,\tau}(\delta) = \ln\!\Big(\frac{64}{\delta}\big(1 + 16 D B_3 \sqrt{n_{i,\tau}}\big)^2\Big).$$

Proof. We decompose the deviation into the (i) full-window block, controlled by a uniform martingale concentration argument, and (ii) the recent-window block, controlled deterministically by a μ-count bound. Throughout, all operator norms are for 2×2 symmetric matrices.

Step 1 (boundedness, Lipschitzness, sparsity of increments). By construction, $\sum_{t\le t_\tau} \Delta f_{i,t}^{\text{full}} = f_{i,\tau}^{\text{full}}$, and the decomposition is $(\mathcal{G}_t)$-adapted because each increment uses only data revealed by time t and the plug-in profile $\widehat{\Psi}^{\tau}$, which is $\mathcal{G}_{t-1}$-measurable.

Sparsity: $\Delta\ell_{i,t}^{\text{purch}} \not\equiv 0$ only if $i \in S_t$, which occurs $n_{i,\tau}$ times. Also $\Delta\ell_{i,t}^{\text{ret,full}} \not\equiv 0$ only if $a_{t-\mu} = i$, and this can happen at most once per period, hence at most $|\mathcal{B}_{i,\tau}^{\text{full}}| \le n_{i,\tau}$ times. The ridge is nonzero only at $t = 1$. Therefore the total number of nonzero increments is at most $2n_{i,\tau} + 1$.

Write $u := v^\top \theta_i = q_i - \rho_i$ and $z := w^\top \theta_i = q_i + \rho_i$. On Θ_i, $u, z \in [-\bar{u}, \bar{u}]$, hence $r(z) \in [r_-, 1 - r_-]$ and $1 - r(z) \in [r_-, 1 - r_-]$.

For purchase terms,

$$-\frac{d^2}{du^2} \log \alpha(u) = \alpha(u)(1 - \alpha(u)) \le \frac{1}{4}, \qquad -\frac{d^2}{du^2} \log(1 - \alpha(u)) = \alpha(u)(1 - \alpha(u)) \le \frac{1}{4}.$$

Since $\|vv^\top\|_{\mathrm{op}} = \|v\|_2^2 = 2$,

$$\left\| -\nabla^2_{\theta_i} \log \alpha(\theta_i) \right\|_{\mathrm{op}} \le \frac{1}{2}, \qquad \left\| -\nabla^2_{\theta_i} \log(1 - \alpha(\theta_i)) \right\|_{\mathrm{op}} \le \frac{1}{2}.$$

For the return term $\log r(z)$,

$$-\frac{d^2}{dz^2} \log r(z) = r(z)(1 - r(z)) \le \frac{1}{4}, \qquad \|ww^\top\|_{\mathrm{op}} = \|w\|_2^2 = 2,$$

so $\| -\nabla^2_{\theta_i} \log r(\theta_i)\|_{\mathrm{op}} \le \frac{1}{2}$.

For the censored term $\log(1 - c\, r(z))$ with $c \in [0,1]$, a direct calculation yields

$$\left| \frac{d^2}{dz^2} \log(1 - c\, r(z)) \right| \le \frac{c\, r(z)(1 - r(z))}{1 - c\, r(z)} + \frac{c^2 r(z)^2 (1 - r(z))^2}{(1 - c\, r(z))^2}.$$

Using $r(z)(1 - r(z)) \le 1/4$ and $1 - c\, r(z) \ge 1 - r(z) \ge r_-$ gives

$$\sup_{z \in [-\bar{u}, \bar{u}],\, c \in [0,1]} \left| \frac{d^2}{dz^2} \log(1 - c\, r(z)) \right| \le \frac{1}{4r_-} + \frac{1}{16 r_-^2},$$

hence

$$\sup_{\theta_i \in \Theta_i,\, c \in [0,1]} \left\| -\nabla^2_{\theta_i} \log(1 - c\, r(\theta_i)) \right\|_{\mathrm{op}} \le 2\left(\frac{1}{4r_-} + \frac{1}{16 r_-^2} \right) = \frac{1}{2r_-} + \frac{1}{8 r_-^2}. \tag{EC.13}$$

The ridge contributes exactly ηI_2 to the negative Hessian. Therefore there is a finite $B_2 = B_2(\bar{u}, \eta)$ bounding $\|\nabla^2_{\theta_i} \Delta f^{\mathrm{full}}_{i,t}(\theta)\|_{\mathrm{op}}$ uniformly.

For Lipschitz, each increment is a finite linear combination of compositions of the form $h(v^\top \theta)$ and $g(w^\top \theta)$ with $h \in \{\log \alpha, \log(1 - \alpha)\}$ and $g \in \{\log r, \log(1 - c\, r) : c \in [0,1]\}$, plus the ridge. On $[-\bar{u}, \bar{u}]$, all third derivatives of these scalar maps are uniformly bounded (with a bound depending only on $\bar{u}$), because $1 - c\, r(z) \ge r_- > 0$ uniformly. For a composition $\theta \mapsto g(w^\top \theta)$,

$$\nabla^2_\theta \big(g(w^\top \theta) \big) = g''(w^\top \theta)\, ww^\top,$$

so by the mean value theorem,

$$\left\| \nabla^2 (g(w^\top \theta)) - \nabla^2 (g(w^\top \theta')) \right\|_{\mathrm{op}} \le \sup |g^{(3)}|\, \|w\|_2^3\, \|\theta - \theta'\|_2,$$

and similarly for $h(v^\top \theta)$ with v. Thus there exists $B_3 = B_3(\bar{u})$ such that for all t and $\theta, \theta' \in \Theta_i$,

$$\left\| \nabla^2_{\theta_i} \Delta f^{\mathrm{full}}_{i,t}(\theta) - \nabla^2_{\theta_i} \Delta f^{\mathrm{full}}_{i,t}(\theta') \right\|_{\mathrm{op}} \le B_3\, \|\theta - \theta'\|_2. \tag{EC.14}$$

Step 2 (uniform concentration for the full block). Fix $\theta \in \Theta_i$ and define the symmetric matrix martingale differences

$$X_t(\theta) := -\nabla^2_{\theta_i}\Delta f^{\text{full}}_{i,t}(\theta) - \mathbb{E}\big[-\nabla^2_{\theta_i}\Delta f^{\text{full}}_{i,t}(\theta)\,\big|\,\mathcal{G}_{t-1}\big], \qquad t = 1, \dots, t_\tau.$$

Then $M_t(\theta) := \sum_{s=1}^t X_s(\theta)$ is a symmetric matrix martingale (w.r.t. $(\mathcal{G}_t)$) and

$$M_{t_\tau}(\theta) = \Big(-\nabla^2_{\theta_i} f^{\text{full}}_{i,\tau}(\theta)\Big) - H^{\text{full}}_{i,\tau}(\theta).$$

By (EC.13), $\|-\nabla^2\Delta f^{\text{full}}_{i,t}(\theta)\|_{\text{op}} \le B_2$ uniformly, hence $\|X_t(\theta)\|_{\text{op}} \le 2B_2 =: R$. Also $X_t(\theta)^2 \preceq \|X_t(\theta)\|^2_{\text{op}} I_2 \preceq R^2 I_2$. By sparsity, at most $2n_{i,\tau}+1$ terms are nonzero, hence the predictable quadratic variation satisfies

$$W_{t_\tau}(\theta) := \sum_{t=1}^{t_\tau} \mathbb{E}[X_t(\theta)^2 \mid \mathcal{G}_{t-1}] \preceq R^2(2n_{i,\tau}+1)\, I_2.$$

Applying the maximal matrix Freedman inequality (Tropp 2011, Thm. 1.2) (dimension 2) to $\pm M_t(\theta)$ and taking a union bound gives: there exists an absolute constant $C > 0$ such that for any $x > 0$, with probability at least $1 - 2e^{-x}$,

$$\sup_{t \le t_\tau} \|M_t(\theta)\|_{\text{op}} \le C\Big(\sqrt{B_2^2(2n_{i,\tau}+1)x} + B_2 x\Big). \tag{EC.15}$$

In particular, (EC.15) holds at time $t = t_\tau$.

Let $\varepsilon_0 := \frac{1}{8B_3\sqrt{n}}$ and let $\mathcal{N}_{\varepsilon_0}$ be an ε_0-net of Θ_i in $\|\cdot\|_2$, with

$$|\mathcal{N}_{\varepsilon_0}| \le \Big(1 + \frac{2D}{\varepsilon_0}\Big)^2 \le \Big(1 + 16DB_3\sqrt{n}\Big)^2.$$

Set

$$x := \ln\Big(\frac{64|\mathcal{N}_{\varepsilon_0}|}{\delta}\Big) \le \ln\Big(\frac{64}{\delta}\big(1 + 16DB_3\sqrt{n}\big)^2\Big) =: \Lambda_{i,\tau}(\delta).$$

Apply (EC.15) with this x and take a union bound over $\theta \in \mathcal{N}_{\varepsilon_0}$. Since $2e^{-x}|\mathcal{N}_{\varepsilon_0}| \le \delta/32$, we obtain that with probability at least $1 - \delta/32$,

$$\sup_{\theta \in \mathcal{N}_{\varepsilon_0}} \|M_{t_\tau}(\theta)\|_{\text{op}} \le C\Big(\sqrt{B_2^2(2n_{i,\tau}+1)\Lambda_{i,\tau}(\delta)} + B_2\Lambda_{i,\tau}(\delta)\Big). \tag{EC.16}$$

Now extend from the net to all of Θ_i. Fix any $\theta \in \Theta_i$ and let $\theta^\sharp \in \mathcal{N}_{\varepsilon_0}$ satisfy $\|\theta - \theta^\sharp\|_2 \le \varepsilon_0$. By (EC.14) and Jensen,

$$\|X_t(\theta) - X_t(\theta^\sharp)\|_{\text{op}} \le \big\|-\nabla^2\Delta f^{\text{full}}_{i,t}(\theta) + \nabla^2\Delta f^{\text{full}}_{i,t}(\theta^\sharp)\big\|_{\text{op}} + \mathbb{E}\Big[\big\|-\nabla^2\Delta f^{\text{full}}_{i,t}(\theta) + \nabla^2\Delta f^{\text{full}}_{i,t}(\theta^\sharp)\big\|_{\text{op}}\,\Big|\,\mathcal{G}_{t-1}\Big] \tag{EC.17}$$

$$\le 2B_3\|\theta - \theta^\sharp\|_2. \tag{EC.18}$$

Summing over nonzero increments (at most $2n_{i,\tau}+1$ of them),

$$\|M_{t_\tau}(\theta) - M_{t_\tau}(\theta^\sharp)\|_{\text{op}} \le \sum_{t \le t_\tau} \|X_t(\theta) - X_t(\theta^\sharp)\|_{\text{op}} \le 2B_3(2n_{i,\tau}+1)\varepsilon_0 \le \frac{1}{2}\sqrt{n\,\Lambda_{i,\tau}(\delta)},$$

where the last inequality uses $\varepsilon_0 = \frac{1}{8B_3\sqrt{n}}$ and $\Lambda_{i,\tau}(\delta) \ge 1$ for $\delta \le e^{-1}$, and otherwise the bound is trivial after adjusting constants. Combining with (EC.16) and absorbing fixed numerical factors into a constant $C_1 = C_1(B_2)$ yields

$$\sup_{\theta_i\in\Theta_i} \left\| \left(-\nabla^2_{\theta_i} f^{\mathrm{full}}_{i,\tau}(\theta_i)\right) - H^{\mathrm{full}}_{i,\tau}(\theta_i) \right\|_{\mathrm{op}} \le C_1\left(\sqrt{n_{i,\tau}\,\Lambda_{i,\tau}(\delta)} + \Lambda_{i,\tau}(\delta)\right)$$

with probability at least $1-\delta$ after enlarging constants.

Step 3 (recent-window block). For each recent purchase $s \in \mathcal{B}^{\mathrm{rec}}_{i,\tau}$, the return contribution in $\ell^{\mathrm{ret,rec}}_{i,\tau}$ is a term of the form $\log r(\theta_i)$ (if $Z^{(m_{s,\tau})}_{i,s} = 1$) or $\log(1 - r(\theta_i)\widehat{\Psi}^\tau(m_{s,\tau}))$ (if $Z^{(m_{s,\tau})}_{i,s} = 0$). By (EC.13) (with $c \in [0,1]$), each such per-purchase term has $-\nabla^2$ operator norm bounded by $\max\{\frac{1}{2}, \frac{1}{2r_-} + \frac{1}{8r_-^2}\}$. There are at most μ such purchases total (at most one purchase per period), hence

$$\sup_{\theta_i\in\Theta_i} \left\|-\nabla^2_{\theta_i}\ell^{\mathrm{ret,rec}}_{i,\tau}(\theta_i)\right\|_{\mathrm{op}} \le \mu\cdot\max\left\{\frac{1}{2},\ \frac{1}{2r_-} + \frac{1}{8r_-^2}\right\} =: C_2\,\mu.$$

Conclusion. Since

$$-\nabla^2_{\theta_i} f_{i,\tau}(\theta_i) = -\nabla^2_{\theta_i} f^{\mathrm{full}}_{i,\tau}(\theta_i) - \nabla^2_{\theta_i}\ell^{\mathrm{ret,rec}}_{i,\tau}(\theta_i),$$

combining Step 2 and Step 3 and taking the supremum over $\theta_i \in \Theta_i$ yields the claimed bound.

□

This uniform Hessian control allows us to lower bound curvature not only at the truth, but uniformly along the segment connecting the estimator and the true parameter.

EC.1.7. Parameter error bound for the per-product estimator

We now combine curvature, self-normalized score concentration, censoring bias, and delay-profile estimation error to obtain a high-probability radius for the per-product estimator used by COBRA.

LEMMA EC.7. *Fix a product i and epoch τ. Define the penalized (censored) per-product objective*

$$f_{i,\tau}(\theta_i) := \ell_{i,\tau}(\theta_i, \widehat{\Psi}) - \frac{\eta}{2}\|\theta_i\|_2^2,$$

and let $\hat\theta_{i,\tau} \in \arg\max_{\theta_i\in\Theta_i} f_{i,\tau}(\theta_i)$.

There exist constants $\kappa_{15}, \kappa_{16} > 0$ such that uniformly over all products $i \in [N]$ and epochs τ, with probability at least $1 - O((NT)^{-2})$, for all $n_{i,\tau} \ge 1$,

$$\left\|\hat\theta_{i,\tau} - \theta_i^*\right\|_\infty \le \frac{\kappa_{15}}{\psi_\mu}\left(\sqrt{\frac{\ln(NT)}{n_{i,\tau}}} + \frac{\min\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\bar{d}\}}{n_{i,\tau}}\right) + \frac{\kappa_\psi}{c_-\,\psi_\mu^2}\max_{m\le\mu}\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right|.$$

Proof. Write $\Delta_i := \hat\theta_{i,\tau} - \theta_i^*$ and $\gamma(s) := \theta_i^* + s\Delta_i$. Let $f := f_{i,\tau}$. Since $\hat\theta_{i,\tau}$ maximizes f over the convex set Θ_i, we have the variational inequality

$$\left\langle \nabla f(\hat\theta_{i,\tau}),\, \theta - \hat\theta_{i,\tau}\right\rangle \le 0 \qquad \forall\theta\in\Theta_i.$$

Taking $\theta = \theta_i^*$ gives $\langle \nabla f(\hat{\theta}_{i,\tau}), \Delta_i \rangle \geq 0$. By the fundamental theorem of calculus,

$$\nabla f(\hat{\theta}_{i,\tau}) - \nabla f(\theta_i^*) = \int_0^1 \nabla^2 f(\gamma(s))\, \Delta_i \, \mathrm{d}s.$$

Taking inner products with Δ_i and rearranging yields

$$\Delta_i^\top \bar{H}_i \Delta_i \leq \left\langle \nabla f(\theta_i^*), \Delta_i \right\rangle, \qquad \bar{H}_i := \int_0^1 \left(-\nabla^2 f(\gamma(s))\right) \mathrm{d}s. \tag{EC.19}$$

Then, we have,

$$\begin{aligned}
\|\Delta_i\|_{\bar{H}_i}^2 &= \Delta_i^\top \bar{H}_i \Delta_i && \text{(by definition)} && \text{(EC.20)}\\
&\leq \left\langle \nabla f(\theta_i^*), \Delta_i \right\rangle && \text{(by (EC.19))} && \text{(EC.21)}\\
&\leq \|\nabla f(\theta_i^*)\|_2 \, \|\Delta_i\|_2 && \text{(Cauchy–Schwarz)} && \text{(EC.22)}\\
&\leq \|\nabla f(\theta_i^*)\|_2 \sqrt{\|\bar{H}_i^{-1}\|_{\mathrm{op}}}\, \|\Delta_i\|_{\bar{H}_i} && \text{(since } \|\Delta_i\|_2^2 \leq \|\bar{H}_i^{-1}\|_{\mathrm{op}} \|\Delta_i\|_{\bar{H}_i}^2) && \text{(EC.23)}
\end{aligned}$$

If $\Delta_i \neq 0$, divide (EC.23) by $\|\Delta_i\|_{\bar{H}_i}$ to get

$$\|\Delta_i\|_{\bar{H}_i} \leq \sqrt{\|\bar{H}_i^{-1}\|_{\mathrm{op}}}\, \|\nabla f(\theta_i^*)\|_2.$$

Finally use $\|\Delta_i\|_2 \leq \sqrt{\|\bar{H}_i^{-1}\|_{\mathrm{op}}}\, \|\Delta_i\|_{\bar{H}_i}$ to conclude

$$\|\Delta_i\|_2 \leq \|\bar{H}_i^{-1}\|_{\mathrm{op}} \, \|\nabla f(\theta_i^*)\|_2.$$

(If $\Delta_i = 0$ the claim is trivial.)

Thus it suffices to lower bound $\lambda_{\min}(\bar{H}_i)$ and upper bound $\|\nabla f(\theta_i^*)\|_2$.

Note

$$-\nabla^2 f(\theta_i) = -\nabla^2_{\theta_i} \ell_{i,\tau}(\theta_i, \widehat{f}_d) + \eta I_2.$$

By Lemma EC.2(ii), for all $\theta_i \in \Theta_i$,

$$H_{i,\tau}^{\mathrm{obs}}(\theta_i, \hat{f}_d) := -\mathbb{E}\left[\nabla^2_{\theta_i} f(\theta_i)\right] \succeq c_- \, \psi_\mu \, n_{i,\tau} \, I_2 \qquad \text{whenever } n_{i,\tau} \geq n_1,$$

for some $n_1 \geq 1$.

Moreover, because f is C^2 on the compact set and each time t reveals only $O(1)$ information about product i, the per-time Hessian increments are uniformly bounded and Lipschitz in θ_i.

Let $\mathcal{E}_{i,\tau}^1$ be the event from Lemma EC.6, we have

$$\sup_{\theta_i \in \Theta_i} \left\| (-\nabla^2_{\theta_i} f(\theta_i)) - H_{i,\tau}^{\mathrm{obs}}(\theta_i) \right\|_{\mathrm{op}} \leq \varepsilon_{i,\tau},$$

where $\varepsilon_{i,\tau} := \widetilde{C}\big(\sqrt{n_{i,\tau}\ln(NT)} + \ln(NT) + \mu\big)$ with probability at least $1-(NT)^{-3}$. Hence, for every $\theta_i \in \Theta_i$,

$$\lambda_{\min}\big(-\nabla^2_{\theta_i} f(\theta_i)\big) \ \ge\ \lambda_{\min}\big(H^{\mathrm{obs}}_{i,\tau}(\theta_i)\big) - \varepsilon_{i,\tau}.$$

If in addition $n_{i,\tau} \ge n_2$ for some $n_2 \ge 1$ so that $\lambda_{\min}(H^{\mathrm{obs}}_{i,\tau}(\theta_i)) \ge c_-\psi_\mu n_{i,\tau}$ uniformly in θ_i and $\varepsilon_{i,\tau} \le \frac{1}{2}c_-\psi_\mu n_{i,\tau}$, then

$$\inf_{\theta_i\in\Theta_i} \lambda_{\min}\big(-\nabla^2_{\theta_i} f(\theta_i)\big) \ \ge\ \tfrac{1}{2}c_-\psi_\mu n_{i,\tau}.$$

Since $\gamma(s) \in \Theta_i$ for all $s \in [0,1]$, integrating over s preserves the PSD order and gives

$$\bar{H}_i = \int_0^1 \Big(-\nabla^2_{\theta_i} f(\gamma(s))\Big)\,\mathrm{d}s \succeq \tfrac{1}{2}c_-\psi_\mu n_{i,\tau} I_2, \qquad \|\bar{H}_i^{-1}\|_{\mathrm{op}} \le \frac{2}{c_-\psi_\mu n_{i,\tau}}.$$

Let $n_0 := \max\{n_1, n_2\}$. For $n_{i,\tau} < n_0$, we always have $\bar{H}_i \succeq \eta I_2$ so invertibility holds; the final rate is trivial after enlarging constants because Θ_i is compact.

We have

$$\nabla f(\theta_i^*) = \nabla_{\theta_i}\ell_{i,\tau}(\theta_i^*, \widehat{f}_d) - \eta\theta_i^*.$$

Decompose

$$\nabla_{\theta_i}\ell_{i,\tau}(\theta_i^*, \widehat{f}_d) = \nabla_{\theta_i}\widetilde{\ell}_{i,\tau}(\theta_i^*, f_d) + b_i, \qquad b_i := \nabla_{\theta_i}\ell_{i,\tau}(\theta_i^*, \widehat{f}_d) - \nabla_{\theta_i}\widetilde{\ell}_{i,\tau}(\theta_i^*, f_d).$$

Write $G_{i,\tau} := \nabla_{\theta_i}\widetilde{\ell}_{i,\tau}(\theta_i^*, f_d)$. Then

$$\|\nabla f(\theta_i^*)\|_2 \le \|G_{i,\tau}\|_2 + \|b_i\|_2 + \eta\|\theta_i^*\|_2.$$

Since $\theta_i^* \in \Theta_i$, $\|\theta_i^*\|_2$ is bounded by an absolute constant depending only on Θ_i and can be absorbed into κ_{15}.

Let $\{\xi_{i,t}\}_{t\le t_\tau}$ denote the $(\mathcal{F}_t)$-adapted per-time score increments for product i in the oracle log-likelihood $\widetilde{\ell}_{i,\tau}(\cdot, f_d)$, so that

$$G_{i,\tau} = \sum_{t=1}^{t_\tau} \xi_{i,t}.$$

Define the active-count process

$$A_{i,t} := \sum_{s=1}^{t} \mathbb{I}\{\xi_{i,s} \ne 0\}, \qquad t \ge 1.$$

With the natural per-time decomposition (purchase reveals at time t and return-related reveals at time t), each exposure of product i (i.e., each time $i \in S_t$) can generate at most one purchase-related nonzero increment, and each purchase of i can generate at most one return-related reveal time. Moreover, purchases can only occur at exposure times. Hence deterministically

$$|\mathcal{B}_{i,\tau}| \le n_{i,\tau}, \qquad A_{i,t_\tau} \le n_{i,\tau} + |\mathcal{B}_{i,\tau}| \le 2n_{i,\tau}.$$

Let $\delta := (NT)^{-3}$ and let $\mathcal{E}^2_{i,\tau}$ be the high-probability event in Lemma EC.3 applied to the martingale $\sum_{t\le t_\tau} \xi_{i,t}$ at time t_τ. Then $\mathbb{P}(\mathcal{E}^2_{i,\tau}) \ge 1-\delta = 1-(NT)^{-3}$, and on $\mathcal{E}^2_{i,\tau}$,

$$\|G_{i,\tau}\|^2_{(V_0^{(i)}+V_{i,\tau}^{(i)})^{-1}} \le \kappa_{\mathrm{det}} \ln(NT),$$

where $V_{i,\tau}^{(i)} := 2H^{(i)}_{*,t_\tau} + 2B_\xi^2 A_{i,t_\tau} I_2$.

By Lemma EC.2(ii) and boundedness on the compact set, $H^{(i)}_{*,t_\tau} \preceq C_+ n_{i,\tau} I_2$, and by the deterministic bound above, $A_{i,t_\tau} \le 2n_{i,\tau}$, hence

$$V_0^{(i)} + V_{i,\tau}^{(i)} \preceq \kappa_V\, n_{i,\tau} I_2 \quad \Rightarrow \quad \lambda_{\max}(V_0^{(i)} + V_{i,\tau}^{(i)}) \le \kappa_V n_{i,\tau}.$$

Therefore

$$\|G_{i,\tau}\|_2^2 \le \lambda_{\max}(V_0^{(i)} + V_{i,\tau}^{(i)})\, \|G_{i,\tau}\|^2_{(V_0^{(i)}+V_{i,\tau}^{(i)})^{-1}} \le \kappa_{16}^2\, n_{i,\tau} \ln(NT),$$

i.e.

$$\|G_{i,\tau}\|_2 \le \kappa_{16}\sqrt{n_{i,\tau}\ln(NT)}.$$

Let $\mathcal{E}^3_{i,\tau}$ be the event in Lemma EC.4. For $\delta := (NT)^{-3}$, $\mathbb{P}(\mathcal{E}^3_{i,\tau}) \ge 1-\delta = 1-(NT)^{-3}$, and on $\mathcal{E}^3_{i,\tau}$,

$$\|b_i\|_2 \le \min\left\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\bar{d}\right\} + \kappa_\psi \psi_\mu^{-1}\, n_{i,\tau} \max_{m\le\mu}\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right|.$$

On the intersection of events $\mathcal{E}^1_{i,\tau}$, $\mathcal{E}^2_{i,\tau}$, and $\mathcal{E}^3_{i,\tau}$, we have

$$\|\Delta_i\|_2 \le \|\bar{H}_i^{-1}\|_{\mathrm{op}}\big(\|G_{i,\tau}\|_2 + \|b_i\|_2 + \eta\|\theta_i^*\|_2\big) \le \frac{2}{c_-\,\psi_\mu\, n_{i,\tau}}\big(\|G_{i,\tau}\|_2 + \|b_i\|_2 + \eta\|\theta_i^*\|_2\big),$$

with probability at least $1-3(NT)^{-3}$ by union bound. Absorbing $\eta\|\theta_i^*\|_2$ into constants yields

$$\|\Delta_i\|_2 \le \frac{\kappa_{15}}{\psi_\mu}\left(\sqrt{\frac{\ln(NT)}{n_{i,\tau}}} + \frac{\min\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)}+\kappa_4\bar{d}\}}{n_{i,\tau}}\right) + \frac{\kappa_\psi}{c_-\,\psi_\mu^2} \max_{m\le\mu}\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right|.$$

Finally $\|\Delta_i\|_\infty \le \|\Delta_i\|_2$, giving the stated bound.

□

Lemma EC.7 provides the product-level confidence radius used in the algorithm and in the regret analysis.

EC.1.8. Proof of Theorem 1

We are now ready to prove the regret bound for COBRA. The argument combines the confidence radii derived above with a UCB-style epoch decomposition of regret.

Proof. Fix the true parameters

$$\theta^* = \{(q_i^*, \rho_i^*)\}_{i=1}^N, \qquad \psi_\mu \in (0,1].$$

Recall that for any assortment $S \subseteq [N]$ with $|S| \le K$, the per-period expected profit under parameters (θ, ψ) is

$$Q(S;\theta,\psi) := \sum_{i\in S}\big(p_i - c_i\psi\, r_i(\theta_i)\big)\frac{\exp(q_i-\rho_i)}{1+\sum_{j\in S}\exp(q_j-\rho_j)}, \qquad r_i(\theta_i) = \frac{1}{1+\exp(q_i+\rho_i)}. \tag{EC.24}$$

Let

$$S^* \in \arg\max_{S:|S|\le K} Q(S;\theta^*,\psi_\mu)$$

be an optimal static assortment for the clairvoyant. Then the regret can be written as

$$\mathrm{Regret}(\pi_{\mathrm{COBRA}}) = \mathbb{E}\left[\sum_{t=1}^{T}\Big(Q(S^*;\theta^*,\psi_\mu) - Q(S_t;\theta^*,\psi_\mu)\Big)\right],$$

where S_t is the assortment offered at time t under COBRA.

In epoch τ, COBRA fixes an assortment S_τ and offers it repeatedly until the first no-purchase. Let $\mathcal{H}_\tau$ be the (random) set of periods in epoch τ, and let $\bar{\tau}$ denote the index of the last epoch. Then $\{1,\ldots,T\} = \bigsqcup_{\tau=1}^{\bar{\tau}}\mathcal{H}_\tau$. Define the epoch-level gap

$$\Delta_\tau := Q(S^*;\theta^*,\psi_\mu) - Q(S_\tau;\theta^*,\psi_\mu).$$

We can rewrite the regret as

$$\mathrm{Regret}(\pi_{\mathrm{COBRA}}) = \mathbb{E}\left[\sum_{\tau=1}^{\bar{\tau}}|\mathcal{H}_\tau|\,\Delta_\tau\right].$$

Fix an epoch τ and its assortment S_τ. Under the true parameters (θ^*, ψ_μ), the no-purchase probability in any period of that epoch is

$$\mathbb{P}(a_t = 0 \mid S_\tau, \theta^*) = \frac{1}{1+\sum_{j\in S_\tau}\exp(q_j^* - \rho_j^*)} =: \frac{1}{D(S_\tau;\theta^*)},$$

where we define

$$D(S;\theta) := 1 + \sum_{j\in S}\exp(q_j-\rho_j).$$

Thus $|\mathcal{H}_\tau|$ is geometric with success probability $1/D(S_\tau;\theta^*)$, and

$$\mathbb{E}\big[|\mathcal{H}_\tau| \mid S_\tau, \theta^*\big] = D(S_\tau;\theta^*). \tag{EC.25}$$

Conditioning on the history just before epoch τ, we obtain

$$\mathrm{Regret}(\pi_{\mathrm{COBRA}}) = \mathbb{E}\left[\sum_{\tau=1}^{\bar{\tau}} D(S_\tau;\theta^*)\,\Delta_\tau\right]. \tag{EC.26}$$

Set $\delta := \frac{1}{N^2T^2}$. Define the event $\mathcal{G}_T$ on which the following hold simultaneously for all epochs $\tau \le \bar{\tau}$ and all products $i \in [N]$:

(a) the per-product estimator satisfies the concentration bound from Lemma EC.7;

(b) the delay CDF estimator satisfies the uniform bound from Lemma EC.5.

By a union bound over (i,τ) and τ (and using $\bar{\tau} \le T$),

$$\mathbb{P}(\mathcal{G}_T^c) \;\le\; (N\bar{\tau})\delta + (\bar{\tau})\delta \;\le\; \frac{2}{NT}. \tag{EC.27}$$

Decompose the regret as

$$\text{Regret}(\pi_{\text{COBRA}}) = R_{\text{good}} + R_{\text{bad}},$$

where

$$R_{\text{good}} := \mathbb{E}\left[\sum_{\tau=1}^{\bar{\tau}} D(S_\tau;\theta^*)\,\Delta_\tau\,\mathbb{I}(\mathcal{G}_T)\right], \qquad R_{\text{bad}} := \mathbb{E}\left[\sum_{\tau=1}^{\bar{\tau}} D(S_\tau;\theta^*)\,\Delta_\tau\,\mathbb{I}(\mathcal{G}_T^c)\right].$$

On the compact parameter set (Assumption 1) and with $|S| \le K$, the function $Q(S;\theta,\psi)$ is uniformly bounded: there exists $Q_{\max} < \infty$ (depending only on $\bar{u}$ and K) such that $|Q(S;\theta,\psi)| \le Q_{\max}$ for all feasible (S,θ,ψ). Hence $|\Delta_\tau| \le 2Q_{\max}$. Also,

$$D(S_\tau;\theta^*) \le 1 + Ke^{\bar{u}} =: D_{\max}, \qquad \bar{\tau} \le T.$$

Therefore,

$$R_{\text{bad}} \;\le\; \mathbb{E}\left[\sum_{\tau=1}^{\bar{\tau}} D_{\max}\cdot 2Q_{\max}\cdot \mathbb{I}(\mathcal{G}_T^c)\right] \;\le\; 2D_{\max}Q_{\max}\,T\,\mathbb{P}(\mathcal{G}_T^c) \;\le\; 4D_{\max}Q_{\max}, \tag{EC.28}$$

which is $O(1)$ and can be absorbed into the final constant C. It remains to bound R_{good}.

On $\mathcal{G}_T$, the true parameters satisfy

$$\theta^* \in C_\tau, \qquad \psi_\mu \in [\underline{\psi}_\tau, 1], \qquad \forall \tau,$$

where C_τ and $\underline{\psi}_\tau$ are the confidence objects used by COBRA. In epoch τ, COBRA chooses

$$S_\tau \in \arg\max_{S:|S|\le K}\left\{\max_{\theta\in C_\tau,\ \psi\in[\underline{\psi}_\tau,1]} Q(S;\theta,\psi)\right\}.$$

Let $(\theta_\tau,\psi_\tau) \in C_\tau \times [\underline{\psi}_\tau, 1]$ be a maximizer for the chosen S_τ. Since $Q(S;\theta,\psi)$ is *nonincreasing in* ψ (it appears only through $-c_i\psi r_i(\theta_i)$), we may take

$$\psi_\tau = \underline{\psi}_\tau. \tag{EC.29}$$

Moreover, for any assortment S (in particular $S = S^*$),

$$\max_{\theta\in C_\tau,\ \psi\in[\underline{\psi}_\tau,1]} Q(S;\theta,\psi) \;\ge\; Q(S;\theta^*,\psi_\mu).$$

Hence, on $\mathcal{G}_T$,

$$Q(S_\tau;\theta_\tau,\psi_\tau) \;\ge\; Q(S^*;\theta^*,\psi_\mu),$$

and therefore

$$\Delta_\tau = Q(S^*;\theta^*,\psi_\mu) - Q(S_\tau;\theta^*,\psi_\mu) \le Q(S_\tau;\theta_\tau,\psi_\tau) - Q(S_\tau;\theta^*,\psi_\mu). \tag{EC.30}$$

Fix an epoch τ and abbreviate S_τ as S. Write $Q(S;\theta,\psi) = N(S;\theta,\psi)/D(S;\theta)$ with

$$N(S;\theta,\psi) := \sum_{i\in S}\big(p_i - c_i\psi\, r_i(\theta_i)\big)\exp(q_i-\rho_i), \qquad D(S;\theta) := 1 + \sum_{j\in S}\exp(q_j-\rho_j).$$

On the compact set, the maps $(q_i,\rho_i)\mapsto \exp(q_i-\rho_i)$ and $(q_i,\rho_i)\mapsto r_i(\theta_i)$ are smooth with uniformly bounded derivatives, and $D(S;\theta)$ is bounded away from 0 and above by a constant depending only on $(\bar u, K)$. Therefore, by the mean-value theorem applied to the scaled difference $D(S;\theta^*)\big(Q(S;\theta,\psi) - Q(S;\theta^*,\psi_\mu)\big)$, there exist finite constants $C_\theta, C_\psi > 0$ depending only on $(\bar u, K)$ such that for all feasible (θ,ψ),

$$D(S;\theta^*)\left|Q(S;\theta,\psi) - Q(S;\theta^*,\psi_\mu)\right| \le C_\theta \sum_{i\in S}\|\theta_i - \theta_i^*\|_\infty + C_\psi\,|\psi-\psi_\mu|. \tag{EC.31}$$

Combining (EC.30) with (EC.31) (and restoring τ) yields

$$D(S_\tau;\theta^*)\,\Delta_\tau \le C_\theta \sum_{i\in S_\tau}\|\theta_{i,\tau} - \theta_i^*\|_\infty + C_\psi\,|\psi_\tau - \psi_\mu|. \tag{EC.32}$$

Let $W_{i,\tau}$ denote a (deterministic) radius such that, on $\mathcal{G}_T$,

$$\|\hat\theta_{i,\tau} - \theta_i^*\|_\infty \le W_{i,\tau} \qquad \text{and} \qquad \theta_{i,\tau}\in C_{i,\tau} \subseteq \{\theta:\ \|\theta - \hat\theta_{i,\tau}\|_\infty \le W_{i,\tau}\}. \tag{EC.33}$$

Then, on $\mathcal{G}_T$,

$$\|\theta_{i,\tau} - \theta_i^*\|_\infty \le \|\theta_{i,\tau} - \hat\theta_{i,\tau}\|_\infty + \|\hat\theta_{i,\tau} - \theta_i^*\|_\infty \le 2W_{i,\tau}. \tag{EC.34}$$

Also, by (EC.29) and the definition of $\underline{\psi}_\tau$, we may choose a radius β_τ such that on $\mathcal{G}_T$,

$$|\psi_\tau - \psi_\mu| = |\underline{\psi}_\tau - \psi_\mu| \le \beta_\tau. \tag{EC.35}$$

Substituting (EC.34)–(EC.35) into (EC.32) gives, on $\mathcal{G}_T$,

$$D(S_\tau;\theta^*)\,\Delta_\tau \le 2C_\theta \sum_{i\in S_\tau} W_{i,\tau} + C_\psi\,\beta_\tau. \tag{EC.36}$$

Using (EC.26) and (EC.36),

$$\begin{aligned}
R_{\text{good}} &= \mathbb{E}\left[\sum_{\tau=1}^{\bar\tau} D(S_\tau;\theta^*)\,\Delta_\tau\,\mathbb{I}(\mathcal{G}_T)\right] \\
&\le \mathbb{E}\left[\sum_{\tau=1}^{\bar\tau}\left(2C_\theta\sum_{i\in S_\tau} W_{i,\tau} + C_\psi\,\beta_\tau\right)\mathbb{I}(\mathcal{G}_T)\right] \\
&\le 2C_\theta\,\mathbb{E}\left[\sum_{\tau=1}^{\bar\tau}\sum_{i\in S_\tau} W_{i,\tau}\right] + C_\psi\,\mathbb{E}\left[\sum_{\tau=1}^{\bar\tau}\beta_\tau\right],
\end{aligned} \tag{EC.37}$$

using $\mathbb{I}(\mathcal{G}_T) \le 1$ and nonnegativity.

On $\mathcal{G}_T$, Lemma EC.7 gives, uniformly over all (i,τ),

$$\left\|\hat{\theta}_{i,\tau} - \theta_i^*\right\|_\infty \le \frac{\kappa_{15}}{\psi_\mu}\left(\sqrt{\frac{\ln(NT)}{n_{i,\tau}}} + \frac{\min\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\bar{d}\}}{n_{i,\tau}}\right) + \frac{\kappa_\psi}{c_-\psi_\mu^2}\max_{m\le\mu}\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right|. \quad \text{(EC.38)}$$

Define β_τ to be the right-hand side bound from Lemma EC.5: with probability at least $1 - O(N^{-2}T^{-2})$,

$$\max_{m\le\mu}\left|\widehat{\Psi}^\tau(m) - \Psi(m)\right| \le \beta_\tau, \quad \text{(EC.39)}$$

where

$$\beta_\tau := c_1\sqrt{\frac{\ln(NT\mu)}{S_\tau^{\text{full}}}} + c_2\frac{\ln(NT\mu)}{S_\tau^{\text{full}}} + \frac{\sum_{(i,s)\in P_\tau^{\text{full}}}\delta_{i,\tau-1}}{S_\tau^{\text{full}}}.$$

Then it is valid to choose

$$W_{i,\tau} := \frac{\kappa_{15}}{\psi_\mu}\left(\sqrt{\frac{\ln(NT)}{n_{i,\tau}}} + \frac{\min\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\bar{d}\}}{n_{i,\tau}}\right) + \frac{\kappa_\psi}{c_-\psi_\mu^2}\beta_\tau. \quad \text{(EC.40)}$$

Fix a product i and let $\mathcal{T}_i := \{\tau \le \bar{\tau} :\ i \in S_\tau\}$ be the epochs in which i is offered; let $n_{i,T} := n_{i,\bar{\tau}}$ be its final exposure count. Ordering $\mathcal{T}_i = \{\tau_1^{(i)} < \cdots < \tau_{m_i}^{(i)}\}$ yields $n_{i,\tau_k^{(i)}} \ge k$, hence

$$\sum_{\tau\in\mathcal{T}_i}\frac{1}{\sqrt{n_{i,\tau}}} \le \sum_{k=1}^{n_{i,T}}\frac{1}{\sqrt{k}} \le 2\sqrt{n_{i,T}}, \quad \text{(EC.41)}$$

$$\sum_{\tau\in\mathcal{T}_i}\frac{1}{n_{i,\tau}} \le 1 + \sum_{k=2}^{n_{i,T}}\frac{1}{k} \le 1 + \ln n_{i,T} \le 1 + \ln T. \quad \text{(EC.42)}$$

Summing (EC.40) over $(\tau, i\in S_\tau)$ and using (EC.41)–(EC.42) gives

$$\begin{aligned}
\sum_{\tau=1}^{\bar{\tau}}\sum_{i\in S_\tau} W_{i,\tau} &= \sum_{i=1}^{N}\sum_{\tau\in\mathcal{T}_i} W_{i,\tau} \\
&\le \frac{\kappa_{15}}{\psi_\mu}\sqrt{\ln(NT)}\sum_{i=1}^{N}2\sqrt{n_{i,T}} + \frac{\kappa_{15}}{\psi_\mu}\min\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\bar{d}\}\sum_{i=1}^{N}(1+\ln T) \\
&\quad + \frac{\kappa_\psi}{c_-\psi_\mu^2}\sum_{\tau=1}^{\bar{\tau}}\sum_{i\in S_\tau}\beta_\tau \\
&\le \frac{2\kappa_{15}}{\psi_\mu}\sqrt{\ln(NT)}\sum_{i=1}^{N}\sqrt{n_{i,T}} + \frac{\kappa_{15}}{\psi_\mu}N(1+\ln T)\min\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\bar{d}\} \\
&\quad + \frac{\kappa_\psi}{c_-\psi_\mu^2}K\sum_{\tau=1}^{\bar{\tau}}\beta_\tau,
\end{aligned} \quad \text{(EC.43)}$$

using $|S_\tau| \le K$.

Since at most K products are offered per period, $\sum_{i=1}^{N} n_{i,T} \le KT$. By Cauchy–Schwarz,

$$\sum_{i=1}^{N}\sqrt{n_{i,T}} \le \sqrt{N\sum_{i=1}^{N} n_{i,T}} \le \sqrt{NKT}.$$

Also absorb κ_3, κ_4 into a single constant $\widetilde{C} > 0$ so that

$$\min\{\mu,\ \kappa_3\sqrt{\mu\ln(NT)} + \kappa_4\overline{d}\} \le \widetilde{C}\cdot\min\{\mu,\ \sqrt{\mu\ln(NT)} + \overline{d}\}.$$

Thus there exist constants $C_1, C_2 > 0$ (depending only on $\bar{u}, K, \eta$) such that

$$\sum_{\tau=1}^{\bar{\tau}}\sum_{i\in S_\tau} W_{i,\tau} \ \le\ \frac{C_1}{\psi_\mu}\sqrt{NKT\ln(NT)} + \frac{C_2}{\psi_\mu}N\ln T\cdot\min\{\mu,\ \sqrt{\mu\ln(NT)} + \overline{d}\} + \frac{C_2}{\psi_\mu^2}K\sum_{\tau=1}^{\bar{\tau}}\beta_\tau. \tag{EC.44}$$

By Lemma EC.5, β_τ has three pieces. The first two are of the form $\sqrt{\ln(NT\mu)/S_\tau^{\text{full}}}$ and $\ln(NT\mu)/S_\tau^{\text{full}}$. Since S_τ^{full} is nondecreasing and $S_{\bar{\tau}}^{\text{full}} \le T$, a standard harmonic-sum argument yields constants $C_3, C_4 > 0$ such that

$$\sum_{\tau=1}^{\bar{\tau}}\left(c_1\sqrt{\frac{\ln(NT\mu)}{S_\tau^{\text{full}}}} + c_2\frac{\ln(NT\mu)}{S_\tau^{\text{full}}}\right) \le C_3\sqrt{T\ln(NT\mu)} + C_4\ln T. \tag{EC.45}$$

For the third (plug-in) term in β_τ, use that $r_i(\theta_i)$ is Lipschitz on the compact set, so the condition $|r_i(\theta_i) - r_i(\hat{\theta}_{i,\tau-1})| \le \delta_{i,\tau-1}$ can be met with $\delta_{i,\tau-1} \le L_r\|\hat{\theta}_{i,\tau-1} - \theta_i^*\|_\infty$ for some $L_r < \infty$. Thus this plug-in contribution can be upper bounded (up to a constant factor) by terms already controlled by the same global sums appearing in (EC.44), and can be absorbed into the constants after enlarging them if needed. Consequently we may write

$$\sum_{\tau=1}^{\bar{\tau}}\beta_\tau \le C_5\sqrt{T\ln(NT\mu)} + C_6\ln T. \tag{EC.46}$$

Combining (EC.37), (EC.44), and (EC.46), we get

$$R_{\text{good}} \le \frac{C_7}{\psi_\mu}\sqrt{NKT\ln(NT)} + \frac{C_8}{\psi_\mu}N\ln T\cdot\min\{\mu,\ \sqrt{\mu\ln(NT)} + \overline{d}\} \tag{EC.47}$$

$$+ \frac{C_9}{\psi_\mu^2}K\Big(\sqrt{T\ln(NT\mu)} + \ln T\Big) + C_{10}\Big(\sqrt{T\ln(NT\mu)} + \ln T\Big). \tag{EC.48}$$

Since $\psi_\mu \in (0,1]$ and $K \ge 1$, the final term is dominated by the $\psi_\mu^{-2}K$ term up to constants, so after enlarging constants we can write

$$R_{\text{good}} \ \le\ \frac{C}{\psi_\mu}\Big(\sqrt{NKT\ln(NT)} + NK\ln T\cdot\min\{\mu,\ \sqrt{\mu\ln(NT)} + \overline{d}\}\Big) \ +\ \frac{CK}{\psi_\mu^2}\Big(\sqrt{T\ln(NT\mu)} + \ln T\Big)$$

for a (larger) constant $C > 0$.

Finally, add the $O(1)$ term R_{bad} from (EC.28) and absorb it into the same C. This proves the theorem. □

EC.2. Efficient Solution of the Optimistic Assortment Problem

PROPOSITION EC.1. *Fix epoch τ, and define*

$$h_{i,\tau}^*(\lambda) = \max_{\theta_i\in C_{i,\tau}} e^{q_i-\rho_i}\Big(p_i - c_i\underline{\psi}_\tau\,\sigma(-(q_i+\rho_i)) - \lambda\Big).$$

Let $g^*_\tau(\lambda)$ *be the sum of the* K *largest positive values among* $\{h^*_{i,\tau}(\lambda)\}_{i=1}^N$. *Then*

$$Z^*_\tau := \max_{S:|S|\le K} \max_{\theta\in C_\tau} Q(S;\theta,\underline{\psi}_\tau)$$

is the unique solution to

$$g^*_\tau(\lambda) = \lambda.$$

Hence Algorithm 2 returns an optimistic assortment.

Proof. For any S,θ, the inequality $Q(S;\theta,\underline{\psi}_\tau) \ge \lambda$ is equivalent to

$$\sum_{i\in S} e^{q_i-\rho_i}\Big(p_i - c_i\underline{\psi}_\tau\,\sigma(-(q_i+\rho_i)) - \lambda\Big) \ge \lambda.$$

Because $C_\tau = \bigotimes_{i=1}^N C_{i,\tau}$, the inner maximization separates across products, so for fixed λ the largest attainable left-hand side is exactly $g^*_\tau(\lambda)$, obtained by selecting the K largest positive values among $\{h^*_{i,\tau}(\lambda)\}_{i=1}^N$. Therefore,

$$Z^*_\tau \ge \lambda \iff g^*_\tau(\lambda) \ge \lambda.$$

For each i, $h^*_{i,\tau}(\lambda)$ is the pointwise maximum of affine decreasing functions of λ, hence is continuous and nonincreasing. Therefore $g^*_\tau(\lambda)$ is also continuous and nonincreasing, so $g^*_\tau(\lambda)-\lambda$ is continuous and strictly decreasing. It follows that the equation $g^*_\tau(\lambda)=\lambda$ has a unique solution, which can be found by binary search, and the corresponding maximizing assortment is given by the K largest positive values of $h^*_{i,\tau}(\lambda)$.

□

EC.3. Extension: product-dependent delay profiles

Our baseline assumes a common delay CDF $\Psi(\cdot)$ across products. In some applications, however, return timing may vary systematically by product category, channel, or fulfillment mode. In some applications, return timing may vary systematically by product category, channel, or fulfillment mode. The model and algorithm extend directly to product-dependent delay profiles by replacing Ψ with $\{\Psi_i\}_{i=1}^N$, where $\Psi_i(m) := \Pr(D \le m \mid \text{return of product } i)$ for $m \in \{0,1,\dots,\mu\}$. All likelihood terms involving the delay distribution are modified by the substitution $\Psi(m) \mapsto \Psi_i(m)$, e.g., censored factors of the form $\log\big(1-r_i(\theta_i)\Psi(m)\big)$ become $\log\big(1-r_i(\theta_i)\Psi_i(m)\big)$. Estimation of each Ψ_i can be carried out nonparametrically via the same CDF-profile approach under right-censoring at μ, and the optimistic step uses the resulting product-wise confidence sets. The tradeoff is statistical: learning Ψ_i separately requires sufficient return observations per product, so the statistical burden scales with the number of distinct delay profiles. A practical compromise is to assume Ψ is shared across $G \ll N$ known groups (e.g., categories or channels), or to parameterize Ψ_i through low-dimensional features, which preserves the same methodology while controlling sample requirements.

EC.4. Lower Bound

Proof of Theorem 2 Let
$$a := 1 - \frac{\psi_\mu}{2}, \qquad b := a + \psi_\mu \epsilon_2.$$
Fix disjoint sets $S^{\text{att}}, S^{\text{ret}} \subseteq [N]$ with $|S^{\text{att}}| = |S^{\text{ret}}| = K/2$, and define the hard instance as in (6). Its optimal assortment is
$$S^\star = S^{\text{att}} \cup S^{\text{ret}}.$$

For any assortment S_t with $|S_t| \le K$, define
$$m_t^{\text{att}} := \frac{K}{2} - |S_t \cap S^{\text{att}}|, \qquad m_t^{\text{ret}} := \frac{K}{2} - |S_t \cap S^{\text{ret}}|, \qquad \ell_t := K - |S_t|,$$
and
$$\delta_t^{\text{att}} := \frac{2m_t^{\text{att}}}{K}, \qquad \delta_t^{\text{ret}} := \frac{2m_t^{\text{ret}}}{K}.$$
A direct calculation gives
$$Q(S^\star) = \frac{a(1+\epsilon_1) + b}{3 + \epsilon_1},$$
and
$$Q(S_t) = \frac{a(1+\epsilon_1) + b - a\left(\epsilon_1 \delta_t^{\text{att}} + \frac{2\ell_t}{K}\right) - \psi_\mu \epsilon_2 \delta_t^{\text{ret}}}{3 + \epsilon_1 - \epsilon_1 \delta_t^{\text{att}} - \frac{2\ell_t}{K}}.$$
Hence
$$\begin{aligned} Q(S^\star) - Q(S_t) &= \frac{\left(\epsilon_1 \delta_t^{\text{att}} + \frac{2\ell_t}{K}\right)(a - \psi_\mu \epsilon_2) + \delta_t^{\text{ret}} \psi_\mu \epsilon_2 (3+\epsilon_1)}{(3+\epsilon_1)\left(3 + \epsilon_1 - \epsilon_1 \delta_t^{\text{att}} - \frac{2\ell_t}{K}\right)} \\ &\ge \frac{\delta_t^{\text{att}} \epsilon_1 \left(1 - \frac{\psi_\mu}{2} - \psi_\mu \epsilon_2\right) + \delta_t^{\text{ret}} \psi_\mu \epsilon_2 (3+\epsilon_1)}{16}, \end{aligned} \tag{EC.49}$$
where we used $\epsilon_1 \le 1/2$, $\epsilon_2 \le 1/4$, and dropped the nonnegative unused-capacity term $2\ell_t/K$.

Now let
$$N_i := \sum_{t=1}^{T} \mathbb{I}\{i \in S_t\}.$$
Summing (EC.49) over t and using
$$\sum_{t=1}^{T} \delta_t^{\text{att}} = T - \frac{2}{K} \sum_{i \in S^{\text{att}}} N_i, \qquad \sum_{t=1}^{T} \delta_t^{\text{ret}} = T - \frac{2}{K} \sum_{i \in S^{\text{ret}}} N_i,$$
we obtain
$$\begin{aligned} \text{Regret} \gtrsim \epsilon_1 \left(1 - \frac{\psi_\mu}{2} - \psi_\mu \epsilon_2\right)\left(T - \frac{1}{K} \sum_{i \in S^{\text{att}}} \mathbb{E}[N_i]\right) \\ + \psi_\mu \epsilon_2 \left(T - \frac{1}{K} \sum_{i \in S^{\text{ret}}} \mathbb{E}[N_i]\right). \end{aligned} \tag{EC.50}$$

We bound the two terms by averaging over a hard family of instances.

Attraction part. Fix S^{ret}, and average over all $S^{\text{att}} \subseteq [N] \setminus S^{\text{ret}}$ with $|S^{\text{att}}| = K/2$. Reindexing as usual,

$$\frac{1}{K} \frac{1}{|\mathcal{A}|} \sum_{S^{\text{att}} \in \mathcal{A}} \sum_{i \in S^{\text{att}}} \mathbb{E}_{S^{\text{att}} \cup S^{\text{ret}}}[N_i]$$

is bounded by a baseline counting term plus a Pinsker term. The counting term is

$$\lesssim \frac{T}{14},$$

using only $\sum_{i=1}^{N} N_i \le TK$ and $K \le N/4$.

For the Pinsker term, compare neighboring instances that differ only in whether a single product i belongs to S^{att}. If P and G denote the induced laws over the full history, then

$$|\mathbb{E}_P[N_i] - \mathbb{E}_G[N_i]| \le T\sqrt{\tfrac{1}{2} D_{\text{KL}}(P \| G)}.$$

Exactly as in Chen and Wang (2018), the one-step KL divergence is $O(\epsilon_1^2/K)$ whenever product i is offered, and zero otherwise. Therefore, by the chain rule,

$$D_{\text{KL}}(P \| G) \lesssim \frac{\epsilon_1^2}{K} \mathbb{E}_P[N_i].$$

Averaging over the hard family and applying Jensen's inequality gives

$$\frac{1}{K} \frac{1}{|\mathcal{A}|} \sum_{S^{\text{att}} \in \mathcal{A}} \sum_{i \in S^{\text{att}}} \mathbb{E}_{S^{\text{att}} \cup S^{\text{ret}}}[N_i] \lesssim \frac{T}{14} + T\sqrt{\frac{\epsilon_1^2 T}{N}}.$$

Substituting this into (EC.50) yields the attraction contribution

$$\gtrsim \epsilon_1 \left(1 - \frac{\psi_\mu}{2} - \psi_\mu \epsilon_2\right)\left(T - T\sqrt{\frac{\epsilon_1^2 T}{N}}\right).$$

Return part. Now fix S^{att}, and average over all $S^{\text{ret}} \subseteq [N] \setminus S^{\text{att}}$ with $|S^{\text{ret}}| = K/2$. The same counting argument gives the baseline term $\lesssim T/14$.

For neighboring instances differing only in whether product i belongs to S^{ret}, the purchase law is identical; only the return observation attached to product i changes. Conditional on offering and purchasing i, the in-window return signal is Bernoulli with means

$$\psi_\mu/2 \qquad \text{and} \qquad \psi_\mu(1/2 - \epsilon_2).$$

Hence the one-step KL divergence is

$$O(\psi_\mu \epsilon_2^2),$$

for example by bounding Bernoulli KL by the corresponding χ^2 distance. Therefore

$$D_{\text{KL}}(P \| Q) \lesssim \psi_\mu \epsilon_2^2 \, \mathbb{E}_P[N_i].$$

Averaging and using Jensen again,

$$\frac{1}{K}\frac{1}{|\mathcal{B}|}\sum_{S^{\text{ret}}\in\mathcal{B}}\sum_{i\in S^{\text{ret}}}\mathbb{E}_{S^{\text{att}}\cup S^{\text{ret}}}[N_i]\lesssim\frac{T}{14}+T\sqrt{\frac{\psi_\mu\epsilon_2^2TK}{N}}.$$

Substituting into (EC.50) yields the return contribution

$$\gtrsim\psi_\mu\epsilon_2\left(T-T\sqrt{\frac{\psi_\mu\epsilon_2^2TK}{N}}\right).$$

Combining the two parts and choosing

$$\epsilon_1\asymp\sqrt{\frac{N}{T}},\qquad\epsilon_2\asymp\sqrt{\frac{N}{TK\psi_\mu}},$$

subject to $\epsilon_1\le 1/2$ and $\epsilon_2\le 1/4$, gives

$$\text{Regret}\ge\Omega\left(\left(1-\frac{\psi_\mu}{2}\right)\sqrt{NT}+\sqrt{\frac{NT\psi_\mu}{K}}\right).$$

This proves the theorem.

□

EC.5. Additional Computational Experiments

We use transaction-level sales and return data from a fast-fashion retailer's online channel over the Fall–Winter season (2018-08 to 2019-03), comprising more than 14 million transactions. We focus on three categories—dresses, cardigans, and leggings—and normalize prices to $[0,1]$. For each SKU, the return cost is taken as the normalized price plus shipping cost. In the real-data experiments, the assortment capacity is set to $K=10$, reflecting the retailer's landing-page display constraint. Table **??** reports summary statistics.

To estimate purchase attractiveness, we assume

$$v_i=\exp(\beta_j p_i),\qquad j\in\{\text{dress},\text{cardigan},\text{leggings}\},$$

where p_i is the normalized price of product i and β_j is category-specific. We estimate β_j by maximizing the regularized log-likelihood

$$\max_\beta\sum_{i\in\mathcal{N}_j}B_i\Big(\beta p_i-\log(1+\exp(\beta p_i))\Big)-\beta^2,$$

where B_i denotes total sales of product i and $\mathcal{N}_j$ is the set of SKUs in category j. The implied purchase probability is

$$q(i)=\frac{\exp(\beta_j p_i)}{1+\exp(\beta_j p_i)}.$$

Table EC.1 Model estimation results.

	Coefficient	Std. Error	z-value	$P > \|z\|$	[95% CI]
β_{dress}	35.0915	0.3709	94.6025	0.0000	[34.3645, 35.8186]
β_{cardigan}	22.4807	0.3746	60.0153	0.0000	[21.7466, 23.2149]
β_{leggings}	34.8906	0.3246	107.4963	0.0000	[34.2545, 35.5268]

Estimated coefficients are reported in Table EC.5. In all three categories, the fitted model strongly outperforms the null model ($\beta = 0$) in likelihood-ratio tests, with pseudo-R^2 values above 0.91.

Product-level return probabilities are estimated empirically from matched sales and returns. To estimate the return-delay distribution, we match purchase and return records and compute the elapsed time between purchase and return. The resulting delay distribution, shown in Figure EC.1, is heavily concentrated in the first two weeks and is well approximated by a geometric law.

Figure EC.2–EC.4 summarize the empirical distributions of the estimated attraction parameters, return probabilities, and prices by category.

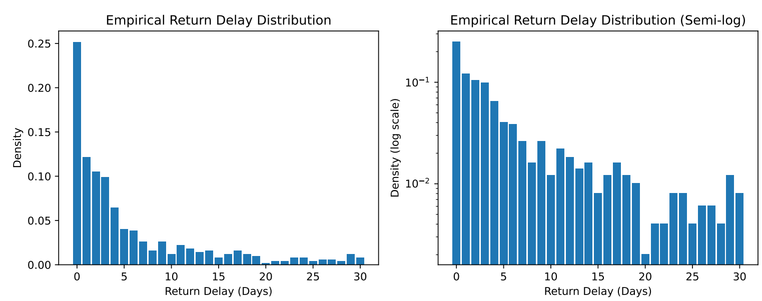

Figure EC.1 **Empirical distribution of return delays pooled across products. The left panel shows the raw distribution, and the right panel shows the same distribution on a semi-log scale. The distribution is strongly right-skewed, and its tail is broadly consistent with exponential decay.**

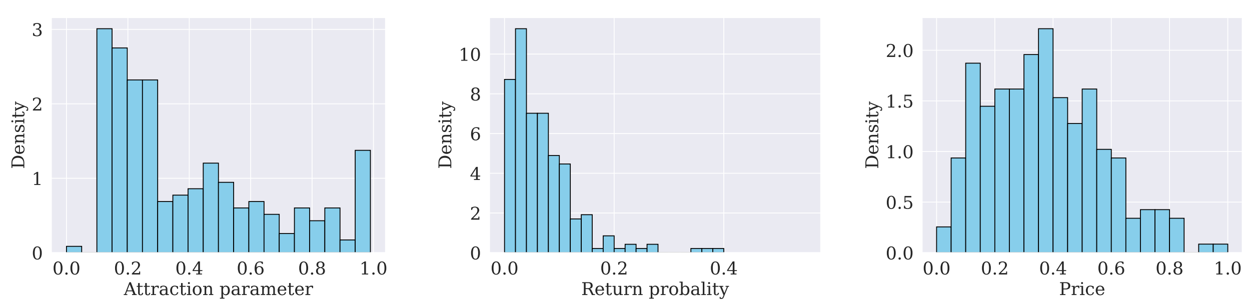

Figure EC.2 **Data distributions for cardigan category: attraction parameters (left), return probabilities (center), and prices (right).**

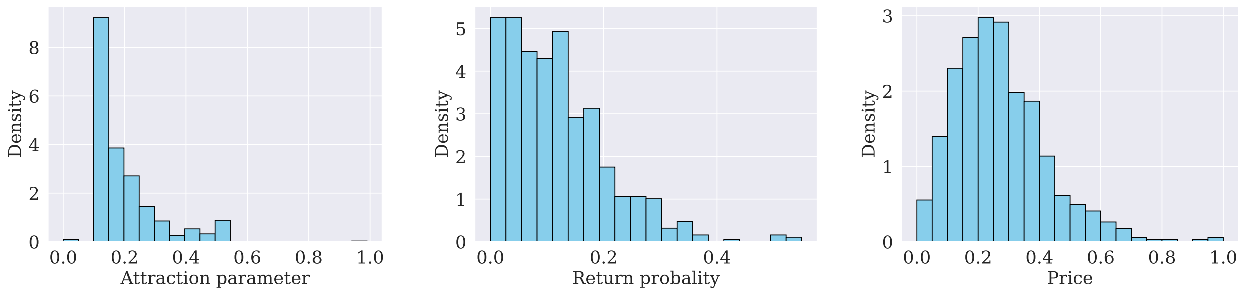

Figure EC.3 Data distributions for dress category: attraction parameters (left), return probabilities (center), and prices (right).

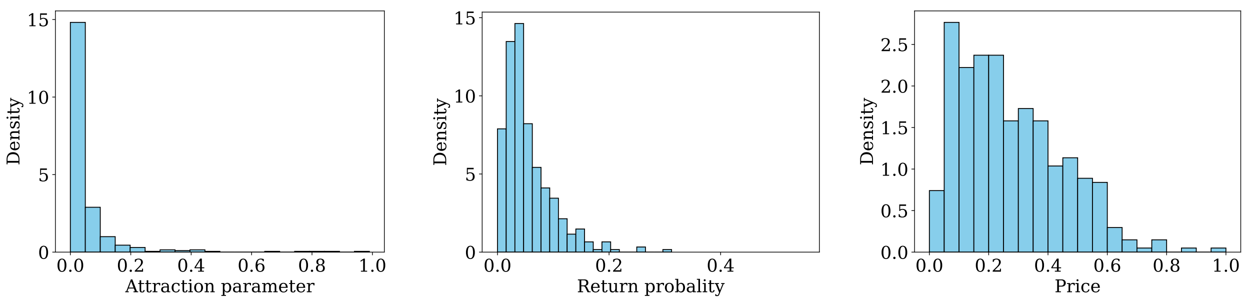

Figure EC.4 Data distributions for leggings category: attraction parameters (left), return probabilities (center), and prices (right).

EC.5.1. Explore-Then-Commit (ETC) Benchmark

As a benchmark, we consider an explore-then-commit policy adapted to our return-aware setting with delayed and window-censored feedback. The policy has two phases. In the exploration phase, it follows a predetermined exploration schedule to collect purchase and return data without performing online optimistic assortment optimization. At the end of exploration, it estimates the product primitives and the in-window delay profile from the collected data, computes the corresponding plug-in optimal assortment, and then commits to this assortment for the remainder of the horizon.